

Unified Nonlinear Control for Position Regulation, Swing Suppression, and Cable-Tautness Guarantees in Quadrotor Suspended-Load Systems

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Abstract

We address the problem of controlling a quadrotor with a cable-suspended load while simultaneously guaranteeing position convergence, cable-swing attenuation, and cable tautness from a single unified design. The key insight is a geometrically exact decomposition of the thrust vector that decouples position regulation from tension regulation at the level of the equations of motion, enabling a principled controller structure that avoids the approximations and singularities common in competing approaches. A Sherman–Morrison precompensator exactly inverts the coupled drone–load inertia, yielding clean closed-loop error dynamics amenable to rigorous Lyapunov analysis. An augmented Lyapunov function certifies local exponential stability of the constant-reference equilibrium, with explicit, verifiable gain conditions and simultaneous exponential bounds on position error, cable swing,

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and cable tension. Numerical simulations on a three-dimensional figure-eight trajectory illustrate the practical trajectory-tracking capability of the proposed controller beyond the constant-reference case covered by the theoretical analysis.

Keywords: Quadrotor UAV, cable-suspended load, swing suppression, cable tautness, Lyapunov stability, exponential stability, geometric control, $SO(3)$

1. Introduction

Quadrotor unmanned aerial vehicles (UAVs) with cable-suspended loads are an attractive solution for aerial transportation because they can move payloads without the weight and mechanical complexity of rigid manipulators [27, 26]. This capability is relevant to package delivery, material transport, infrastructure inspection, and search-and-rescue missions in complex or GPS-denied environments [35, 41, 42, 37, 38]. As these applications move toward more autonomous and demanding operation, recent advances in urban air mobility management, soft aerial vehicles, and novel multi-rotor configurations have pushed aerial robotics beyond conventional quadrotor stabilization and trajectory-tracking problems [36, 39, 40, 1, 6, 3, 2, 4, 5]. In the suspended-load case, this increased level of autonomy is complicated by the payload pendulum dynamics, which strongly couple the vehicle and load motions. Therefore, the controller must regulate or track the vehicle position while simultaneously suppressing cable swing and preserving positive cable tension throughout the maneuver.

The system considered in this paper is shown in Fig. 1. A quadrotor of mass m_q transports a point-mass load m_L through a massless inexten-

19 sible cable of length ℓ . The cable direction is represented by $\mathbf{q} \in \mathbb{S}^2$, the
 20 cable tension by λ , the quadrotor position by $\mathbf{p}_q$, and the load position by
 21 $\mathbf{p}_L = \mathbf{p}_q + \ell \mathbf{q}$. Since a cable can only pull, the condition $\lambda \geq 0$ is a physical
 22 feasibility requirement. If the tension vanishes, the cable becomes slack and
 23 the pendulum model no longer describes the system. Therefore, cable taut-
 24 ness is not only a safety condition, but also a necessary condition for swing
 25 suppression.

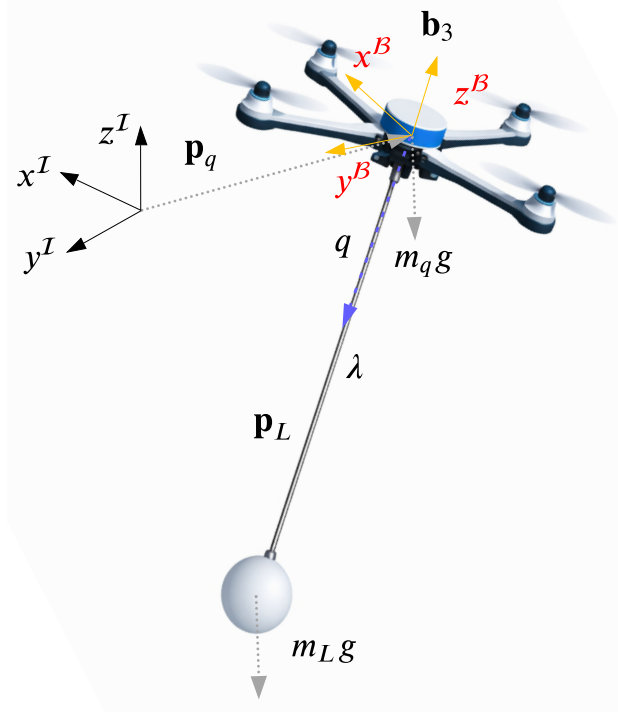


Figure 1: Quadrotor UAV with cable-suspended load. The thrust direction is $\mathbf{b}_3 = R\mathbf{e}_3$,
 the cable direction is $\mathbf{q} \in \mathbb{S}^2$, the tension is $\lambda \geq 0$, and the load position is $\mathbf{p}_L = \mathbf{p}_q + \ell \mathbf{q}$.

26 *Related work*

27 Research on UAVs with suspended loads has been approached from both
 28 planning and feedback control perspectives. Early planning results exploited

the differential flatness of the taut-cable model to generate load-aware trajectories with reduced swing excitation [8, 9]. This line was later extended to mixed-integer and time-optimal formulations [10, 11], and more recent work combines motion planning with geometric tracking controllers validated on real platforms [33]. These methods are useful for generating feasible motions, but they do not by themselves provide closed-loop swing rejection when disturbances or modeling errors move the system away from the planned trajectory.

Feedback control has therefore received considerable attention. Lee [12] introduced geometric controllers for quadrotors with suspended loads using the cable direction on $\mathbb{S}^2$, and later extensions considered rigid-body payloads [13]. The attitude loop commonly used in these designs builds on the geometric controller on $\text{SO}(3)$ proposed in [7]. Other Lyapunov-based and passivity-based approaches include the exponential and adaptive controllers of Yang and Xian [14, 15], the hierarchical energy-coupling schemes of Liang et al. [16, 17], and the trajectory-tracking framework of Cabecinhas, Cunha, and Silvestre [29]. Robustness and performance have also been addressed using active disturbance rejection [30], adaptive backstepping with unknown load mass [31], global path-following and set-stabilization methods [21, 22], finite-time control under wind disturbances [19], nonlinear swing damping [18], elastic-cable adaptive control [20], uncertain-mass geometric adaptation [23], neural-network-based passivity control [24], and high-performance slung-load control under external disturbances [32]. Comprehensive surveys are available in [27, 26, 25].

Contributions

Table 1 provides a qualitative comparison with representative methods from the literature; it highlights that the proposed controller combines an exact tangential–radial thrust decomposition with a local Lyapunov proof that explicitly certifies cable tautness through the bound $\lambda(t) \geq m_L g/2 > 0$. To the best of our knowledge, prior works typically assume tautness as part of the model validity domain, whereas the present work provides an explicit local closed-loop tautness bound as a formal consequence of the control law.

Table 1: Qualitative comparison with representative suspended-load UAV control methods. The symbol ‘ $\sim$ ’ indicates that tautness is assumed as part of the model validity domain, but not certified as a closed-loop property.

Reference	Tautness certified	Tangential–radial split	SO(3) inner loop	Euler-angle free	Lyapunov proof	Explicit tautness bound
Lee [12], <i>TCST</i> 2014	$\sim$	—	✓	✓	✓	—
Yang & Xian [14], <i>TCST</i> 2020	$\sim$	—	—	—	✓	—
Liang et al. [16], <i>TIE</i> 2018	$\sim$	—	—	—	✓	—
Cabecinhas et al. [29], <i>Automatica</i> 2019	$\sim$	—	—	—	✓	—
Akhtar & Waslander [21], <i>TCST</i> 2022	$\sim$	—	✓	✓	✓	—
Yang et al. [19], <i>TH</i> 2024	$\sim$	—	—	—	✓	—
Kong et al. [31], <i>Automatica</i> 2024	$\sim$	—	—	—	✓	—
This paper	✓	✓	✓	✓	✓	✓

Despite the progress above, most existing controllers either assume cable tautness or verify it after the controller has been designed. This paper addresses this limitation by exploiting an exact geometric decomposition of the thrust vector into two components: a tangential component that drives position tracking and a radial component that regulates cable tension. This de-

composition follows directly from the constraint $\mathbf{q} \in \mathbb{S}^2$ and does not rely on small-angle approximations or linearization. The resulting controller achieves position regulation, swing suppression, and positive cable tension within a single Lyapunov-based design.

The main contributions are the following:

- (i) A complete 24-state model of the quadrotor-load system, including the translational dynamics, the cable dynamics on $\mathbb{S}^2$, and the full attitude dynamics on $\text{SO}(3)$. A Sherman–Morrison inversion [34] gives an explicit expression for the quadrotor acceleration without configuration-dependent matrix inversions.
- (ii) An exact tangential–radial decomposition of the virtual thrust vector (Proposition 1). This decomposition separates the position and tension channels at the level of the nonlinear equations of motion, not as an approximation.
- (iii) A Lyapunov-based outer-loop controller (Theorem 1) that guarantees local exponential *regulation* for constant references, simultaneous swing suppression, and cable tautness, with explicit gain conditions. The same controller is numerically shown to perform well in smooth trajectory-tracking tasks beyond the constant-reference case.
- (iv) An integration with a geometric inner-loop attitude controller on $\text{SO}(3)$ [7], together with a complete numerical validation on a three-dimensional figure-eight trajectory.

The rest of the paper is organized as follows. Section 2 presents the model and the control objectives. Section 3 develops the control law and

90 the stability analysis. Section 4 reports the simulation results, and Section 5
 91 concludes the paper.

92 *Notation.* Bold lower-case letters denote column vectors in $\mathbb{R}^3$; upper-case
 93 letters denote matrices. $I \in \mathbb{R}^{3 \times 3}$ is the identity. The hat map $(\hat{\cdot}) : \mathbb{R}^3 \rightarrow$
 94 $\mathfrak{so}(3)$ satisfies $\hat{x}y = x \times y$. The vee map $(\cdot)^\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$ is its inverse. $\|\cdot\|$
 95 is the Euclidean norm. $P_T(\mathbf{q}) = I - \mathbf{q}\mathbf{q}^\top$ denotes the tangential projector
 96 onto the plane orthogonal to $\mathbf{q}$.

97 2. System Modeling and Problem Formulation

98 2.1. Configuration, kinematics, and assumptions

99 Consider a quadrotor of mass m_q carrying a point-mass load of mass m_L
 100 suspended by a cable of length $\ell > 0$. We work in an inertial frame with
 101 gravity $g > 0$ directed along $-\mathbf{e}_3$, where $\mathbf{e}_3 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^\top$. The following
 102 assumptions are maintained throughout.

103 **Assumption 1** (Ideal cable and taut-cable model). *The cable is massless*
 104 *and inextensible, and it can transmit only tensile forces along its own axis.*
 105 *The spherical-pendulum model used below is valid on the taut-cable domain*
 106 *($\lambda \geq 0$). One of the control objectives is therefore to certify that the closed-*
 107 *loop trajectories remain in this domain.*

108 **Assumption 2** (Anchor point). *The cable is attached at the center of mass*
 109 *of the drone. Consequently, the cable tension produces no net torque on the*
 110 *drone.*

111 **Assumption 3** (Quadrotor model). *The quadrotor attitude is described by*
 112 *$R \in \text{SO}(3)$; the body angular velocity in body coordinates is $\boldsymbol{\Omega} \in \mathbb{R}^3$. The*

113 total thrust $T > 0$ acts along the body z -axis $\mathbf{b}_3 := R\mathbf{e}_3$, and the control
 114 torques are $\boldsymbol{\tau} \in \mathbb{R}^3$. The inertia tensor $J = J^\top \succ 0$.

115 2.2. Translational and rotational dynamics

116 Let $\mathbf{p}_q \in \mathbb{R}^3$ and $\mathbf{p}_L \in \mathbb{R}^3$ denote the positions of the drone and the load,
 117 respectively, and define the unit cable direction vector

$$\mathbf{p}_L = \mathbf{p}_q + \ell \mathbf{q}, \quad \mathbf{q} \in \mathbb{R}^3, \quad \|\mathbf{q}\| = 1, \quad (1)$$

118 where $\mathbf{q}$ points from the drone to the load. The unit-norm constraint means
 119 that the load moves on a sphere of radius ℓ centered at $\mathbf{p}_q$. The quadrotor
 120 position and attitude kinematics are

$$\dot{\mathbf{p}}_q = \mathbf{v}_q, \quad \dot{R} = R\hat{\Omega}. \quad (2)$$

121 Differentiating (1) twice (with ℓ constant):

$$\ddot{\mathbf{p}}_L = \ddot{\mathbf{p}}_q + \ell \ddot{\mathbf{q}}. \quad (3)$$

122 *Rotational dynamics.* Under Assumption 2, the Euler equations for the drone
 123 read

$$J\dot{\Omega} = \boldsymbol{\tau} - \Omega \times (J\Omega). \quad (4)$$

124 *Translational dynamics.* Let $\lambda \geq 0$ be the scalar cable tension. The equations
 125 of translational motion for the drone and load are

$$m_q \ddot{\mathbf{p}}_q = T\mathbf{b}_3 - m_q g \mathbf{e}_3 + \lambda \mathbf{q}, \quad (5)$$

$$m_L \ddot{\mathbf{p}}_L = -m_L g \mathbf{e}_3 - \lambda \mathbf{q} + \mathbf{f}_{\text{drag}}. \quad (6)$$

126 The sign convention is consistent with Newton's third law: the cable pulls
 127 the drone *downward* along $\mathbf{q}$ (since $\mathbf{q}$ points from the drone to the load) and

pulls the load *upward* along $-\mathbf{q}$. The term $\mathbf{f}_{\text{drag}}$ models aerodynamic drag on the load; its structure is specified in Section 2.3 below.

2.3. Tension, decoupling, and cable dynamics

Definition 1 (Tangential projector). *For any unit vector $\mathbf{q} \in \mathbb{R}^3$, the tangential projector is*

$$P_T(\mathbf{q}) := I - \mathbf{q}\mathbf{q}^\top \in \mathbb{R}^{3 \times 3}. \quad (7)$$

P_T is the orthogonal projector onto the tangent space $T_{\mathbf{q}}\mathbb{S}^2 = \{\boldsymbol{\xi} \in \mathbb{R}^3 : \mathbf{q}^\top \boldsymbol{\xi} = 0\}$. Its basic properties are summarised below.

Lemma 1 (Properties of P_T). *For all unit $\mathbf{q}$ and all $\mathbf{w} \in \mathbb{R}^3$:*

- (a) Symmetry and idempotency: $P_T^\top = P_T$, $P_T^2 = P_T$.
- (b) Kernel and image: $\ker P_T = \text{span}\{\mathbf{q}\}$, $\text{im } P_T = T_{\mathbf{q}}\mathbb{S}^2$.
- (c) Orthogonality: $\mathbf{q}^\top P_T \mathbf{w} = 0$, $P_T \mathbf{q} = \mathbf{0}$.
- (d) Spectral decomposition: $\mathbf{w} = \underbrace{(\mathbf{q}\mathbf{q}^\top)\mathbf{w}}_{\text{radial}} + \underbrace{P_T \mathbf{w}}_{\text{tangential}}$, with $\sigma(P_T) = \{1, 1, 0\}$.

Proof. All properties follow directly from Definition (7) and $\|\mathbf{q}\| = 1$. $\square$

Drag model. Two cases are considered.

Tangential drag (standard model). If drag acts only in the plane perpendicular to the cable,

$$\mathbf{f}_{\text{drag}} = -c_t P_T(\dot{\mathbf{p}}_q + \ell \dot{\mathbf{q}}) = -c_t P_T \dot{\mathbf{p}}_L, \quad c_t \geq 0. \quad (8)$$

By Lemma 1(c), $\mathbf{q}^\top \mathbf{f}_{\text{drag}} = 0$; drag then introduces dissipation only in the pendulum dynamics (Section 2.3) and does not affect the tension formula.

146 *Drag with axial component.* If $\mathbf{f}_{\text{drag}}$ has a non-zero projection on $\mathbf{q}$ (e.g., axial
 147 downwash), then $\mathbf{q}^\top \mathbf{f}_{\text{drag}} \neq 0$ and the tension is modified accordingly; see
 148 (12) below.

149 *Cable tension.* Since λ acts only along $\mathbf{q}$ (Assumption 1), it is isolated by
 150 pre-multiplying (6) by $\mathbf{q}^\top$:

$$m_L \mathbf{q}^\top (\ddot{\mathbf{p}}_L + g \mathbf{e}_3) = -\lambda + \mathbf{q}^\top \mathbf{f}_{\text{drag}}. \quad (9)$$

151 Using the unit-norm constraint $\|\mathbf{q}\|^2 = 1$, differentiated twice:

$$\mathbf{q}^\top \dot{\mathbf{q}} = 0, \quad \mathbf{q}^\top \ddot{\mathbf{q}} = -\|\dot{\mathbf{q}}\|^2. \quad (10)$$

152 Pre-multiplying (3) by $\mathbf{q}^\top$ and applying (10):

$$\mathbf{q}^\top \ddot{\mathbf{p}}_L = \mathbf{q}^\top \ddot{\mathbf{p}}_q + \ell \mathbf{q}^\top \ddot{\mathbf{q}} = \mathbf{q}^\top \ddot{\mathbf{p}}_q - \ell \|\dot{\mathbf{q}}\|^2. \quad (11)$$

153 Substituting and solving for λ gives the *explicit tension formula*:

$$\boxed{\lambda = -m_L [\mathbf{q}^\top (\ddot{\mathbf{p}}_q + g \mathbf{e}_3) - \ell \|\dot{\mathbf{q}}\|^2] + \mathbf{q}^\top \mathbf{f}_{\text{drag}}.} \quad (12)$$

154 **Remark 1** (Centripetal contribution). *The term $m_L \ell \|\dot{\mathbf{q}}\|^2$ in (12) is the*
 155 *centripetal (inward-radial) reaction of the rotating cable and contributes pos-*
 156 *itively to the tension, reducing the risk of slack. The taut-cable validity con-*
 157 *dition $\lambda \geq 0$ must be monitored during operation.*

158 *Drone acceleration via Sherman–Morrison.* Substituting (12) into (5) and
 159 using tangential drag ($\mathbf{q}^\top \mathbf{f}_{\text{drag}} = 0$), the translational equation of the drone
 160 becomes

$$(m_q I + m_L \mathbf{q} \mathbf{q}^\top) \ddot{\mathbf{p}}_q = T \mathbf{b}_3 - m_q g \mathbf{e}_3 - m_L (\mathbf{q}^\top g \mathbf{e}_3 - \ell \|\dot{\mathbf{q}}\|^2) \mathbf{q}. \quad (13)$$

161 Define the *mass ratio* $\mu := \frac{m_L}{m_q}$ and divide by m_q :

$$(I + \mu \mathbf{q} \mathbf{q}^\top) \ddot{\mathbf{p}}_q = \frac{T}{m_q} \mathbf{b}_3 - g \mathbf{e}_3 - \mu (\mathbf{q}^\top g \mathbf{e}_3 - \ell \|\dot{\mathbf{q}}\|^2) \mathbf{q}. \quad (14)$$

162 The matrix $I + \mu \mathbf{q} \mathbf{q}^\top$ is rank-one perturbed and its inverse follows from the
 163 Sherman–Morrison formula:

$$(I + \mu \mathbf{q} \mathbf{q}^\top)^{-1} = I - \frac{\mu}{1 + \mu} \mathbf{q} \mathbf{q}^\top. \quad (15)$$

164 Applying (15) to (14) and using $(I - \frac{\mu}{1+\mu} \mathbf{q} \mathbf{q}^\top) \mathbf{q} = \frac{1}{1+\mu} \mathbf{q}$:

$$\ddot{\mathbf{p}}_q = \left(I - \frac{\mu}{1 + \mu} \mathbf{q} \mathbf{q}^\top \right) \left(\frac{T}{m_q} \mathbf{b}_3 - g \mathbf{e}_3 \right) - \frac{\mu}{1 + \mu} (\mathbf{q}^\top g \mathbf{e}_3 - \ell \|\dot{\mathbf{q}}\|^2) \mathbf{q}, \quad \mu := \frac{m_L}{m_q}.$$

(16)

165 **Remark 2** (Physical interpretation). *The operator $I - \frac{\mu}{1+\mu} \mathbf{q} \mathbf{q}^\top$ reduces the*
 166 *effective thrust in the cable direction by a factor of $\frac{1}{1+\mu}$, since part of the*
 167 *drone thrust is used to support the load. In the tangential directions the full*
 168 *thrust acts; in the radial direction it is attenuated.*

169 *Cable dynamics.* Defining $\mathbf{s} := \dot{\mathbf{q}}$ and pre-multiplying (6) by P_T (using $P_T \mathbf{q} =$
 170 $\mathbf{0}$ to eliminate λ) yields the *tangential* component of $\ddot{\mathbf{q}}$:

$$\ell P_T \ddot{\mathbf{q}} = -P_T (\ddot{\mathbf{p}}_q + g \mathbf{e}_3) - \frac{c_t}{m_L} P_T (\mathbf{v}_q + \ell \mathbf{s}). \quad (17)$$

171 The *radial* component follows from the constraint $\|\mathbf{q}\|^2 = 1$:

$$\ell (\mathbf{q} \mathbf{q}^\top) \ddot{\mathbf{q}} = -\ell \|\mathbf{s}\|^2 \mathbf{q}. \quad (18)$$

172 Adding (17) and (18) gives the full cable equation:

$$\ell \ddot{\mathbf{q}} = -P_T (\ddot{\mathbf{p}}_q + g \mathbf{e}_3) - \frac{c_t}{m_L} P_T (\mathbf{v}_q + \ell \mathbf{s}) - \ell \|\mathbf{s}\|^2 \mathbf{q},$$

(19)

173 which is the equation of a *spherical pendulum with a moving pivot at $\mathbf{p}_q$* and
 174 damping c_t . The last term is the centripetal acceleration directed inward
 175 along the cable. The algebraic constraints

$$\mathbf{q}^\top \mathbf{s} = 0, \quad \mathbf{q}^\top \dot{\mathbf{s}} = -\|\mathbf{s}\|^2, \quad (20)$$

176 are maintained by construction and must be enforced numerically (e.g., by
 177 projecting $\mathbf{s} \leftarrow P_T \mathbf{s}$ at each time step).

178 2.4. State-space summary, objectives, and error variables

179 Define the state $x := (\mathbf{p}_q, \mathbf{v}_q, R, \boldsymbol{\Omega}, \mathbf{q}, \mathbf{s})$, inputs $u = (T, \boldsymbol{\tau})$, and param-
 180 eters $(m_q, m_L, \ell, J, g, c_t)$. The complete model reads:

$$\dot{\mathbf{p}}_q = \mathbf{v}_q, \quad (21a)$$

$$\dot{\mathbf{v}}_q = \left(I - \frac{\mu}{1+\mu} \mathbf{q} \mathbf{q}^\top \right) \left(\frac{T}{m_q} \mathbf{b}_3 - g \mathbf{e}_3 \right) - \frac{\mu}{1+\mu} (\mathbf{q}^\top g \mathbf{e}_3 - \ell \|\mathbf{s}\|^2) \mathbf{q}, \quad (21b)$$

$$\dot{R} = R \hat{\boldsymbol{\Omega}}, \quad (21c)$$

$$J \dot{\boldsymbol{\Omega}} = \boldsymbol{\tau} - \boldsymbol{\Omega} \times (J \boldsymbol{\Omega}), \quad (21d)$$

$$\dot{\mathbf{q}} = \mathbf{s}, \quad (21e)$$

$$\dot{\mathbf{s}} = -\frac{1}{\ell} P_T (\dot{\mathbf{v}}_q + g \mathbf{e}_3) - \frac{c_t}{m_L \ell} P_T (\mathbf{v}_q + \ell \mathbf{s}) - \|\mathbf{s}\|^2 \mathbf{q}, \quad (21f)$$

181 where $\dot{\mathbf{v}}_q$ in (21f) is given by (21b), and $\mathbf{b}_3 = R \mathbf{e}_3$.

182 **Remark 3** (Algebraic constraints). *Equation (21) is a DAE on the manifold*
 183 $\mathbb{S}^2 \times T\mathbb{S}^2 \times \text{SO}(3) \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3$. *The constraints (20) restrict $(\mathbf{q}, \mathbf{s})$ to the*
 184 *tangent bundle $T\mathbb{S}^2$ and must be preserved by numerical integration.*

185 Let $\mathbf{p}_d : [0, \infty) \rightarrow \mathbb{R}^3$ be a desired position trajectory for the drone. The
 186 primary control objectives are:

187 (O1) **Position regulation/practical tracking:** for a constant reference
 188 $\mathbf{p}_d^*, \mathbf{p}_q(t) \rightarrow \mathbf{p}_d^*$ as $t \rightarrow \infty$ (formal guarantee). For smooth time-varying
 189 references, the controller is evaluated as a practical tracking law in
 190 simulation.

191 (O2) **Oscillation suppression:** whenever $\mathbf{p}_d$ becomes constant, the cable
 192 converges to its natural hanging equilibrium, $\mathbf{q}(t) \rightarrow -\mathbf{e}_3$ and $\mathbf{s}(t) \rightarrow \mathbf{0}$.

193 (O3) **Cable tautness:** $\lambda(t) \geq \lambda_{\min} > 0$ for all $t \geq 0$ within the certified
 194 local domain, where $\lambda_{\min} = m_L g/2$ is established by Theorem 1.

195 Objective (O2) is the key physical requirement: any residual cable swing
 196 is undesirable regardless of whether the drone has reached $\mathbf{p}_d$. Objective (O3)
 197 is a feasibility constraint: if tension drops to zero the cable dynamics become
 198 ill-posed and (O2) cannot be achieved.

199 **Remark 4** (On Assumption 1 and tautness). *Assumption 1 is not used as*
 200 *a closed-loop tautness guarantee. It specifies the physical domain in which*
 201 *the cable-suspended-load model is valid: the spherical-pendulum equations of*
 202 *motion hold only when $\lambda \geq 0$. Objective (O3), and later Theorem 1, certify*
 203 *that the proposed controller preserves this model-validity condition locally by*
 204 *ensuring $\lambda(t) \geq m_L g/2 > 0$ along the closed-loop trajectories. Hence, no*
 205 *circularity arises: the assumption defines the domain; the theorem proves*
 206 *invariance of that domain.*

207 *Tension as mechanism, not objective.* The objectives above make no explicit
 208 mention of a tension reference. Nevertheless, the tension is *actively shaped*
 209 through the radial component of the thrust vector: the control law assigns

210 the radial thrust to support the total system weight $(1 + \mu)g$ at equilibrium,
 211 while an additional term $-K_c P_T(\mathbf{q})\mathbf{s}$ injects tangential damping into the
 212 pendulum directly through the thrust direction. This simultaneously enforces
 213 cable tautness ((O3)) and damps pendulum oscillations ((O2)). No separate
 214 tension reference is needed; the hover tension $\lambda = m_L g$ is a *consequence* of
 215 the correct equilibrium, not a separate design choice. This is the central
 216 design idea of the paper.

217 *Error variables.* Define the position and velocity tracking errors

$$\mathbf{e}_p := \mathbf{p}_q - \mathbf{p}_d, \quad \mathbf{e}_v := \mathbf{v}_q - \dot{\mathbf{p}}_d. \quad (22)$$

218 The closed-loop equilibrium consistent with all three objectives is

$$(\mathbf{e}_p, \mathbf{e}_v, \mathbf{q}, \mathbf{s}) = (\mathbf{0}, \mathbf{0}, -\mathbf{e}_3, \mathbf{0}), \quad (23)$$

219 i.e., the drone reaches the desired position, the cable hangs vertically, and all
 220 velocities are zero. At this equilibrium the tension equals $\lambda = m_L g$ (verified
 221 in Lemma 2).

222 *Cascade structure.* The control is organized in two nested loops.

- 223 • **Outer loop:** Design the virtual thrust vector $\mathbf{u} := \frac{T}{m_q} \mathbf{b}_3 \in \mathbb{R}^3$ (mag-
 224 nitude T/m_q , direction $\mathbf{b}_3$) to achieve (O1)–(O2).
- 225 • **Inner loop:** Given $\mathbf{u}^*$, extract $T^* = m_q \|\mathbf{u}^*\|$ and the desired body
 226 z -axis $\mathbf{b}_3^* = \mathbf{u}^* / \|\mathbf{u}^*\|$, then design $\boldsymbol{\tau}$ so that $\mathbf{b}_3 \rightarrow \mathbf{b}_3^*$ exponentially.

227 The cascade analysis follows standard arguments (see, e.g., [7, 28]): if the
 228 inner loop is exponentially stable and the outer loop is exponentially stable

when $\mathbf{b}_3 = \mathbf{b}_3^*$, then the full cascade is exponentially stable. We focus here on the outer-loop design.

Assumption 4 (Perfect attitude tracking). *For the outer-loop stability analysis we assume $\mathbf{b}_3 = \mathbf{b}_3^*$, i.e., $\mathbf{u} = \frac{T}{m_q} \mathbf{b}_3^* = \mathbf{u}^*$ exactly. The attitude error enters as a bounded perturbation in the cascade and is treated separately.*

3. Control Design and Stability Analysis

The control architecture, illustrated in Fig. 2, is a two-time-scale cascade structured around the geometric decoupling identified in Section 2. The *outer loop* (Section 3.1) operates on the translational and cable states $(\mathbf{e}_p, \mathbf{e}_v, \mathbf{q}, \mathbf{s})$ and produces a virtual thrust vector $\mathbf{u}^*$ that simultaneously drives the position error to zero and injects damping into the pendulum mode. A *thrust-allocation* step extracts the desired thrust magnitude T^* and attitude R_d from $\mathbf{u}^*$. The *inner loop* (Section 3.2) is a bounded geometric controller on $\text{SO}(3)$ that tracks R_d exponentially, decoupled from the outer loop by a time-scale separation argument. An online *tension monitor* computes $\hat{\lambda}$ from available measurements, providing a continuous tautness certificate (Objective O3). Formal exponential stability of the full cascade is established in Section 3.3.

3.1. Decoupling and outer-loop control law

The main results of this section are two-fold. Proposition 1 establishes an exact geometric decomposition of the virtual thrust vector into independent position and tension channels directly at the level of the equations of motion. Theorem 1 then uses this structure to design a Lyapunov-based outer-loop

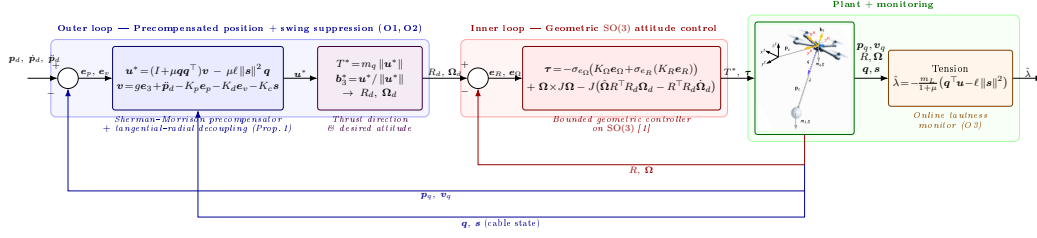


Figure 2: Cascade control architecture. *Blue*: outer loop — precompensated position control and swing suppression (Objectives O1–O2). *Red*: inner geometric SO(3) attitude loop. *Green*: plant and online tension monitor (Objective O3). Feedback signals: $(\mathbf{p}_q, \mathbf{v}_q)$ to the position summing junction, $(\mathbf{R}, \boldsymbol{\Omega})$ to the attitude summing junction, and $(\mathbf{q}, \mathbf{s})$ directly to the outer-loop controller.

252 controller that simultaneously certifies position regulation, cable-swing sup-
 253 pression, and cable tautness with a single analytical certificate.

254 Define the virtual control $\mathbf{u} := \frac{T}{m_q} \mathbf{b}_3$. Substituting (16) into (12) and
 255 using the identity $\mathbf{q}^\top (I - \frac{\mu}{1+\mu} \mathbf{q} \mathbf{q}^\top) = \frac{1}{1+\mu} \mathbf{q}^\top$ with cancellation of the gravity
 256 terms yields

$$\boxed{\lambda = -\frac{m_L}{1+\mu} (\mathbf{q}^\top \mathbf{u} - \ell \|\mathbf{s}\|^2)}. \quad (24)$$

257 This relation shows that λ depends *only* on the radial component $\mathbf{q}^\top \mathbf{u}$ and
 258 the centripetal term $\ell \|\mathbf{s}\|^2$; the tangential component $P_T \mathbf{u}$ does not appear,
 259 confirming the geometric decoupling.

260 **Proposition 1** (Tangential–radial decoupling). *For any virtual control $\mathbf{u}$,*
 261 *the decomposition $\mathbf{u} = P_T \mathbf{u} + (\mathbf{q}^\top \mathbf{u}) \mathbf{q}$ satisfies:*

262 (a) *The cable tension λ depends only on $\mathbf{q}^\top \mathbf{u}$ (radial channel), not on $P_T \mathbf{u}$.*

263 (b) *The tangential drone acceleration $P_T \ddot{\mathbf{p}}_q$ depends only on $P_T \mathbf{u}$ (tangen-*
 264 *tial channel), not on $\mathbf{q}^\top \mathbf{u}$.*

Hence position tracking and tension regulation are geometrically decoupled control channels.

Proof. Part (a) is immediate from (24) and Lemma 1(c). Part (b) follows from applying P_T to (16) and noting that $P_T \mathbf{q} = \mathbf{0}$. $\square$

The key structural insight is that the correct hover equilibrium for the virtual control is $\mathbf{u}_{\text{eq}} = (1 + \mu)g\mathbf{e}_3$, not $g\mathbf{e}_3$: the drone must support its own weight *and* the load weight through the cable. We design the outer-loop law around this equilibrium as follows,

$$\mathbf{u} = \underbrace{(I + \mu \mathbf{q} \mathbf{q}^\top)}_{\Sigma(\mathbf{q})^{-1}} \underbrace{(g\mathbf{e}_3 + \ddot{\mathbf{p}}_d - K_p \mathbf{e}_p - K_d \mathbf{e}_v - K_c \mathbf{s})}_{\text{desired acceleration + gravity}} - \underbrace{\mu \ell \|\mathbf{s}\|^2 \mathbf{q}}_{\text{centripetal compensation}}, \quad (25)$$

with gains $K_p, K_d, K_c > 0$. The physical commands are

$$T^* = m_q \|\mathbf{u}\|, \quad \mathbf{b}_3^* = \frac{\mathbf{u}}{\|\mathbf{u}\|}. \quad (26)$$

Remark 5 (Design insights). *Four decisions determine the structure of (25). (i) At equilibrium $(I + \mu \mathbf{q} \mathbf{q}^\top)|_{\mathbf{q}=-\mathbf{e}_3} \cdot g\mathbf{e}_3 = (1 + \mu)g\mathbf{e}_3$, recovering the correct hover thrust and $\lambda = m_L g > 0$. (ii) The precompensator $(I + \mu \mathbf{q} \mathbf{q}^\top) = \Sigma(\mathbf{q})^{-1}$ exactly inverts the Sherman–Morrison operator acting on the position-tracking command, so that the closed-loop error dynamics are free of projection residuals (see (27)). (iii) The term $-K_c \mathbf{s}$ inside the precompensator is not projected through $P_T(\mathbf{q})$ on purpose: since $\mathbf{q}^\top \mathbf{s} = 0$ exactly, the projection is redundant, and leaving it unprojected keeps the formula compact. (iv) The centripetal term $-\mu \ell \|\mathbf{s}\|^2 \mathbf{q}$ cancels the nonlinear centripetal forcing that would otherwise act as a radial residual in $\dot{\mathbf{e}}_v$.*

284 *Equilibrium verification.* At $(\mathbf{e}_p, \mathbf{e}_v, \mathbf{q}, \mathbf{s}) = (\mathbf{0}, \mathbf{0}, -\mathbf{e}_3, \mathbf{0})$ with $\ddot{\mathbf{p}}_d = \mathbf{0}$:

$$\mathbf{u} = (I + \mu \mathbf{e}_3 \mathbf{e}_3^\top) g \mathbf{e}_3 - \mathbf{0} = (1 + \mu) g \mathbf{e}_3$$

285 so $\mathbf{q}^\top \mathbf{u} = -(1 + \mu)g$ and $\lambda = -\frac{m_L}{1+\mu}(-(1 + \mu)g - 0) = m_L g > 0$.

286 The drone acceleration vanishes:

$$\ddot{\mathbf{p}}_q = \Sigma(-\mathbf{e}_3)((1 + \mu)g \mathbf{e}_3 - g \mathbf{e}_3) - \mu \ell \cdot \mathbf{0} \cdot (-\mathbf{e}_3) = \Sigma(-\mathbf{e}_3) \mu g \mathbf{e}_3,$$

287 since

$$\Sigma(-\mathbf{e}_3) \mu g \mathbf{e}_3 = (I - \frac{\mu}{1 + \mu} \mathbf{e}_3 \mathbf{e}_3^\top) \mu g \mathbf{e}_3 = \mu g \mathbf{e}_3 - \frac{\mu^2 g}{1 + \mu} \mathbf{e}_3 = \frac{\mu g}{1 + \mu} \mathbf{e}_3$$

288 and the centripetal gravity term

$$-\frac{\mu}{1 + \mu} (q^\top g \mathbf{e}_3 - \ell \|s\|^2) q = -\frac{\mu}{1 + \mu} (-g - 0)(-\mathbf{e}_3) = -\frac{\mu g}{1 + \mu} \mathbf{e}_3$$

289 cancels it exactly, giving $\ddot{\mathbf{p}}_q = \mathbf{0}$.

290 **Lemma 2** (Equilibrium and tension). *Under control law (25) with $\ddot{\mathbf{p}}_d = \mathbf{0}$,*
 291 *at the equilibrium (23):*

292 (a) $\ddot{\mathbf{p}}_q = \mathbf{0}$ (drone is stationary);

293 (b) $\lambda = m_L g > 0$ (cable taut at hover tension);

294 (c) $\ddot{\mathbf{q}} = \mathbf{0}$ (cable at rest).

295 *Proof.* (a) and (b) shown above. (c) From (19) with $\ddot{\mathbf{p}}_q = \mathbf{0}$, $\mathbf{s} = \mathbf{0}$:

$$\ell \ddot{\mathbf{q}} = -P_T(-\mathbf{e}_3)(g \mathbf{e}_3) - \mathbf{0} = -(I - \mathbf{e}_3 \mathbf{e}_3^\top)(g \mathbf{e}_3) = -g \mathbf{e}_3 + g \mathbf{e}_3 = \mathbf{0}$$

296

□

297 *Closed-loop acceleration.* Substituting (25) into (16) and using

$$\Sigma(\mathbf{q})(I + \mu\mathbf{q}\mathbf{q}^\top) = I$$

298 it follows that,

$$\begin{aligned} \ddot{\mathbf{p}}_q &= \Sigma(\mathbf{q})[(I + \mu\mathbf{q}\mathbf{q}^\top)(g\mathbf{e}_3 + \ddot{\mathbf{p}}_d - K_p\mathbf{e}_p - K_d\mathbf{e}_v - K_c\mathbf{s}) - \mu\ell\|\mathbf{s}\|^2\mathbf{q} - g\mathbf{e}_3] \\ &\quad - \frac{\mu}{1+\mu}(\mathbf{q}^\top g\mathbf{e}_3 - \ell\|\mathbf{s}\|^2)\mathbf{q} \\ &= \ddot{\mathbf{p}}_d - K_p\mathbf{e}_p - K_d\mathbf{e}_v - K_c\mathbf{s} \\ &\quad + \underbrace{\Sigma(\mathbf{q})(I + \mu\mathbf{q}\mathbf{q}^\top)g\mathbf{e}_3 - \Sigma(\mathbf{q})g\mathbf{e}_3 - \mu\ell\|\mathbf{s}\|^2\Sigma(\mathbf{q})\mathbf{q} - \frac{\mu}{1+\mu}(\mathbf{q}^\top g\mathbf{e}_3)\mathbf{q} + \frac{\mu\ell}{1+\mu}\|\mathbf{s}\|^2\mathbf{q}}_{=0}, \end{aligned}$$

299 where the underbraced remainder vanishes because $\Sigma(I + \mu\mathbf{q}\mathbf{q}^\top)g\mathbf{e}_3 = g\mathbf{e}_3 =$
 300 $\Sigma g\mathbf{e}_3 + \frac{\mu}{1+\mu}(\mathbf{q}^\top g\mathbf{e}_3)\mathbf{q}$ and $\Sigma\mathbf{q} = \frac{1}{1+\mu}\mathbf{q}$. The precompensator $\Sigma(\mathbf{q})^{-1} = I +$
 301 $\mu\mathbf{q}\mathbf{q}^\top$ exactly inverts the Sherman–Morrison operator acting on the drone
 302 acceleration, so (27) directly gives the *exact*, residual-free closed-loop error
 303 dynamics:

$$\dot{\mathbf{e}}_p = \mathbf{e}_v, \tag{27a}$$

$$\dot{\mathbf{e}}_v = -K_p\mathbf{e}_p - K_d\mathbf{e}_v - K_c\mathbf{s}. \tag{27b}$$

304 There are no projection residuals $\mathbf{r}_1, \mathbf{r}_2$, no centripetal remainder $\mathbf{r}_3$, and no
 305 reference-feedforward leakage $\mathbf{r}_4$: the velocity error evolves as a linear PD
 306 system driven solely by the cable angular velocity $\mathbf{s}$, valid for any $\mathbf{q} \in \mathbb{S}^2$ and
 307 any bounded reference trajectory (Assumption 5).

308 3.2. Inner-loop attitude control and cascade structure

309 Given $\mathbf{u}^*$ from (25), the desired thrust magnitude and body z -axis are

$$T^* = m_q \|\mathbf{u}^*\|, \quad \mathbf{b}_3^* = \frac{\mathbf{u}^*}{\|\mathbf{u}^*\|}. \tag{28}$$

is constructed as follows. Choose an auxiliary reference vector $\mathbf{b}_{1c} = \mathbf{e}_1$ if $|\mathbf{b}_3^{*\top} \mathbf{e}_1| < 0.9$, otherwise $\mathbf{b}_{1c} = \mathbf{e}_2$ (to avoid singularity). Then

$$\mathbf{b}_{2d} = \frac{\mathbf{b}_3^* \times \mathbf{b}_{1c}}{\|\mathbf{b}_3^* \times \mathbf{b}_{1c}\|}, \quad \mathbf{b}_{1d} = \mathbf{b}_{2d} \times \mathbf{b}_3^*, \quad R_d = \begin{bmatrix} \mathbf{b}_{1d} & \mathbf{b}_{2d} & \mathbf{b}_3^* \end{bmatrix}. \quad (29)$$

The desired angular velocity $\boldsymbol{\Omega}_d$ is obtained by differentiating R_d : $\hat{\boldsymbol{\Omega}}_d = R_d^\top \dot{R}_d$, computed numerically from the time history of $R_d(t)$.

For the inner-loop torque we adopt the bounded geometric attitude controller developed by the authors in [2]. Defining the attitude and angular-velocity errors

$$\mathbf{e}_R = \frac{1}{2\sqrt{1 + \text{tr}(R_d^\top R)}} (R_d^\top R - R^\top R_d)^\vee \in \mathbb{R}^3, \quad \mathbf{e}_\Omega = \boldsymbol{\Omega} - R^\top R_d \hat{\boldsymbol{\Omega}}_d \in \mathbb{R}^3, \quad (30)$$

the control torque is given by

$$\boldsymbol{\tau} = -\sigma_{e_\Omega}(K_\Omega \mathbf{e}_\Omega + \sigma_{e_R}(K_R \mathbf{e}_R)) + \boldsymbol{\Omega} \times J\boldsymbol{\Omega} - J(\hat{\boldsymbol{\Omega}} R^\top R_d \hat{\boldsymbol{\Omega}}_d - R^\top R_d \dot{\hat{\boldsymbol{\Omega}}}_d), \quad (31)$$

where $K_R, K_\Omega > 0$ are gain matrices and $\sigma_{e_R}, \sigma_{e_\Omega}$ are smooth linear saturation functions with parameters (L_{e_R}, M_{e_R}) and $(L_{e_\Omega}, M_{e_\Omega})$, respectively [2]. This controller guarantees exponential stability of $(\mathbf{e}_R, \mathbf{e}_\Omega) = (\mathbf{0}, \mathbf{0})$ with bounded torques, provided the hemisphere condition $\Psi(R, R_d) = 2 - \sqrt{1 + \text{tr}(R_d^\top R)} < 2$ holds initially (see [2], Theorem 1).

3.3. Stability analysis

Assumption 5 (Constant position reference). *The desired quadrotor position is constant:*

$$\mathbf{p}_d(t) \equiv \mathbf{p}_d^*, \quad \dot{\mathbf{p}}_d(t) = \mathbf{0}, \quad \ddot{\mathbf{p}}_d(t) = \mathbf{0}, \quad \forall t \geq 0.$$

326 **Theorem 1** (Local exponential stability). *Under Assumptions 1–5 and the*
 327 *control law (25)–(31), suppose that the gains and the parameter $\eta > 0$ satisfy*
 328

$$\sigma \in \left(0, \min\{\sqrt{K_p}, K_d\}\right), \quad (32a)$$

$$A := M \begin{bmatrix} K_d - \sigma & \frac{\sigma K_d}{2} \\ \frac{\sigma K_d}{2} & \sigma K_p \end{bmatrix} \succ 0, \quad M := m_q + m_L, \quad (32b)$$

$$\gamma_s := c_t \ell^2 - m_L \ell K_c > 0, \quad (32c)$$

$$\beta_s := \gamma_s - \eta m_L \ell^2 - \frac{M^2 K_c^2 (1 + \sigma^2) + m_L^2 \ell^2 K_p^2 + \ell^2 (m_L K_d - c_t)^2}{\lambda_{\min}(A)} > 0, \quad (32d)$$

$$\beta_0 := \min\left\{\frac{\lambda_{\min}(A)}{2}, \beta_s, \frac{\eta m_L g \ell}{2}\right\}, \quad (32e)$$

$$0 < \eta < \sqrt{\frac{g}{\ell}}, \quad (32f)$$

$$\beta_0 > \eta m_L \ell^2 C_R, \quad (32g)$$

329 where $C_R > 0$ is the local remainder bound satisfying $|\tilde{R}_c| \leq C_R \|z\|^2$ in
 330 a neighbourhood of the equilibrium. Then the equilibrium $(\mathbf{e}_p, \mathbf{e}_v, \mathbf{q}, \mathbf{s}) =$
 331 $(\mathbf{0}, \mathbf{0}, -\mathbf{e}_3, \mathbf{0})$ is locally exponentially stable. More precisely, there exist $r, \rho, c_1, c_2 >$
 332 0 such that for all $V(0) < r$:

333 (i) (Position settling, O1): $\|\mathbf{e}_p(t)\| + \|\mathbf{e}_v(t)\| \leq c_1 e^{-\rho t};$

334 (ii) (Oscillation suppression, O2): $\|\mathbf{q}(t) + \mathbf{e}_3\| + \|\mathbf{s}(t)\| \leq c_2 e^{-\rho t};$

335 (iii) (Tautness, O3): $\lambda(t) \geq \frac{m_L g}{2} > 0$ for all $t \geq 0$.

336 *Proof.* Throughout the proof, the notation $f(z) = O(\|z\|^k)$ means that there

337 exist constants $C > 0$ and $\delta > 0$ such that

$$|f(z)| \leq C \|z\|^k, \quad (33)$$

338 for all z in a neighbourhood $\mathcal{U}$ of the equilibrium, where

$$z := (\mathbf{e}_p, \mathbf{e}_v, \mathbf{q} + \mathbf{e}_3, \mathbf{s}). \quad (34)$$

339 Different occurrences of $O(\|z\|^k)$ may correspond to different constants C ,
 340 which can be combined into a single bound when needed.

341 Under Assumption 5, $\dot{\mathbf{p}}_d = \ddot{\mathbf{p}}_d = \mathbf{0}$, so $\mathbf{e}_v = \mathbf{v}_q$ and the exact error
 342 dynamics (27) become

$$\dot{\mathbf{e}}_p = \mathbf{e}_v, \quad (35a)$$

$$\dot{\mathbf{e}}_v = -K_p \mathbf{e}_p - K_d \mathbf{e}_v - K_c \mathbf{s}. \quad (35b)$$

343 **Step 1: Lyapunov candidate.** We perform the Lyapunov analysis in a
 344 neighbourhood of the equilibrium

$$(\mathbf{e}_p, \mathbf{e}_v, \mathbf{q}, \mathbf{s}) = (\mathbf{0}, \mathbf{0}, -\mathbf{e}_3, \mathbf{0}), \quad (36)$$

345 with $(\mathbf{q}, \mathbf{s}) \in TS^2$, where

$$TS^2 = \{(\mathbf{q}, \mathbf{s}) \in S^2 \times \mathbb{R}^3 : \mathbf{q}^\top \mathbf{s} = 0\}. \quad (37)$$

346 Define the error state

$$z := (\mathbf{e}_p, \mathbf{e}_v, \mathbf{q} + \mathbf{e}_3, \mathbf{s}), \quad (38)$$

347 and consider the augmented Lyapunov function

$$V := V_1 + V_2 + V_c, \quad (39)$$

348 with

$$V_1 := \frac{M}{2} \begin{bmatrix} \mathbf{e}_p \\ \mathbf{e}_v \end{bmatrix}^\top \begin{bmatrix} K_p I & \sigma I \\ \sigma I & I \end{bmatrix} \begin{bmatrix} \mathbf{e}_p \\ \mathbf{e}_v \end{bmatrix} = \frac{M}{2} \|\mathbf{e}_v\|^2 + \frac{MK_p}{2} \|\mathbf{e}_p\|^2 + M\sigma \mathbf{e}_p^\top \mathbf{e}_v, \quad (40)$$

$$V_2 := \frac{m_L g \ell}{2} \|\mathbf{q} + \mathbf{e}_3\|^2 + \frac{m_L \ell^2}{2} \|\mathbf{s}\|^2 = m_L g \ell (1 + \mathbf{q}^\top \mathbf{e}_3) + \frac{m_L \ell^2}{2} \|\mathbf{s}\|^2, \quad (41)$$

$$V_c := \eta m_L \ell^2 (\mathbf{q} + \mathbf{e}_3)^\top \mathbf{s}. \quad (42)$$

349 V_1 is positive definite in $(\mathbf{e}_p, \mathbf{e}_v)$ since $\sigma < \sqrt{K_p}$. V_2 is the mechanical energy
350 of the spherical pendulum with $V_2 \geq 0$ and $V_2 = 0 \Leftrightarrow (\mathbf{q}, \mathbf{s}) = (-\mathbf{e}_3, \mathbf{0})$. V_c
351 is a cross term coupling the cable deflection to its rate.

352 The sum $V_2 + V_c$ is positive definite in $(\mathbf{q} + \mathbf{e}_3, \mathbf{s})$ near $\mathbf{q} = -\mathbf{e}_3$. Using
353 the identity $1 + \mathbf{q}^\top \mathbf{e}_3 = \frac{1}{2} \|\mathbf{q} + \mathbf{e}_3\|^2$ on S^2 and defining $\boldsymbol{\xi} := \mathbf{q} + \mathbf{e}_3$, the sum
354 $V_2 + V_c$ can be written as the quadratic form

$$V_2 + V_c = \frac{1}{2} \begin{bmatrix} \boldsymbol{\xi} \\ \mathbf{s} \end{bmatrix}^\top \begin{bmatrix} m_L g \ell I_3 & \eta m_L \ell^2 I_3 \\ \eta m_L \ell^2 I_3 & m_L \ell^2 I_3 \end{bmatrix} \begin{bmatrix} \boldsymbol{\xi} \\ \mathbf{s} \end{bmatrix}. \quad (43)$$

355 The associated 2×2 block has determinant $m_L^2 \ell^3 (g - \eta^2 \ell)/4 > 0$, which is
356 ensured by $\eta < \sqrt{g/\ell}$ in (32f). Hence, $V_2 + V_c$ is locally positive definite near
357 $(\mathbf{q}, \mathbf{s}) = (-\mathbf{e}_3, \mathbf{0})$.

358 Hence there exist $\alpha_1, \alpha_2 > 0$ and a neighbourhood $\mathcal{U}$ of the equilibrium
359 such that

$$\alpha_1 \|z\|^2 \leq V \leq \alpha_2 \|z\|^2, \quad z := (\mathbf{e}_p, \mathbf{e}_v, \mathbf{q} + \mathbf{e}_3, \mathbf{s}), \quad \forall z \in \mathcal{U}. \quad (44)$$

360 **Step 2:** $\dot{V}_1$. Differentiating (40) along (35) gives

$$\begin{aligned} \dot{V}_1 &= M \mathbf{e}_v^\top \dot{\mathbf{e}}_v + M K_p \mathbf{e}_p^\top \dot{\mathbf{e}}_p + M \sigma \dot{\mathbf{e}}_p^\top \mathbf{e}_v + M \sigma \mathbf{e}_p^\top \dot{\mathbf{e}}_v \\ &= -M(K_d - \sigma) \|\mathbf{e}_v\|^2 - M \sigma K_p \|\mathbf{e}_p\|^2 - M \sigma K_d \mathbf{e}_p^\top \mathbf{e}_v - M K_c \mathbf{e}_v^\top \mathbf{s} - M \sigma K_c \mathbf{e}_p^\top \mathbf{s}. \end{aligned} \quad (45)$$

361 Equivalently,

$$\dot{V}_1 = -(e_v, e_p)^\top A(e_v, e_p) - MK_c(e_v + \sigma e_p)^\top s, \quad (46)$$

362 where from (32b):

$$A := M \begin{bmatrix} K_d - \sigma & \frac{\sigma K_d}{2} \\ \frac{\sigma K_d}{2} & \sigma K_p \end{bmatrix} \succ 0. \quad (47)$$

363 Therefore,

$$\dot{V}_1 \leq -\lambda_{\min}(A)(\|e_v\|^2 + \|e_p\|^2) + MK_c\|e_v\|\|s\| + M\sigma K_c\|e_p\|\|s\|. \quad (48)$$

364 **Step 3:** $\dot{V}_2$. Differentiating (41) gives

$$\dot{V}_2 = m_L g \ell \dot{\mathbf{q}}^\top \mathbf{e}_3 + m_L \ell^2 \mathbf{s}^\top \dot{\mathbf{s}}. \quad (49)$$

365 Since $\dot{\mathbf{q}} = \mathbf{s}$, this becomes

$$\dot{V}_2 = m_L g \ell \mathbf{s}^\top \mathbf{e}_3 + m_L \ell^2 \mathbf{s}^\top \dot{\mathbf{s}}. \quad (50)$$

366 Substituting the cable dynamics (19) yields

$$\dot{V}_2 = m_L g \ell \mathbf{s}^\top \mathbf{e}_3 - m_L \ell \mathbf{s}^\top P_T(\mathbf{q})(\dot{\mathbf{v}}_q + g\mathbf{e}_3) - c_t \ell \mathbf{s}^\top P_T(\mathbf{q})(\mathbf{v}_q + \ell \mathbf{s}) - m_L \ell^2 \|\mathbf{s}\|^2 \mathbf{s}^\top \mathbf{q}. \quad (51)$$

367 Using $P_T(\mathbf{q})\mathbf{s} = \mathbf{s}$ and $\mathbf{q}^\top \mathbf{s} = 0$, we obtain

$$\dot{V}_2 = m_L g \ell \mathbf{s}^\top \mathbf{e}_3 - m_L \ell \mathbf{s}^\top (\dot{\mathbf{v}}_q + g\mathbf{e}_3) - c_t \ell \mathbf{s}^\top (\mathbf{v}_q + \ell \mathbf{s}). \quad (52)$$

368 The gravity terms cancel exactly, giving

$$\dot{V}_2 = -m_L \ell \mathbf{s}^\top \dot{\mathbf{v}}_q - c_t \ell \mathbf{s}^\top \mathbf{v}_q - c_t \ell^2 \|\mathbf{s}\|^2. \quad (53)$$

369 Substituting $\dot{\mathbf{v}}_q = -K_p \mathbf{e}_p - K_d \mathbf{e}_v - K_c \mathbf{s}$ and $\mathbf{e}_v = \mathbf{v}_q$ yields

$$\dot{V}_2 = m_L \ell K_p \mathbf{e}_p^\top \mathbf{s} + \ell (m_L K_d - c_t) \mathbf{e}_v^\top \mathbf{s} - \gamma_s \|\mathbf{s}\|^2, \quad (54)$$

370 where

$$\gamma_s := c_t \ell^2 - m_L \ell K_c > 0 \quad (55)$$

371 by (32c).

372 **Step 4:** $\dot{V}_c$. Differentiating (42) and using $\dot{\mathbf{q}} = \mathbf{s}$:

$$\dot{V}_c = \eta m_L \ell^2 \left[\underbrace{\|\mathbf{s}\|^2}_{\dot{\mathbf{q}}^\top \mathbf{s}} + (\mathbf{q} + \mathbf{e}_3)^\top \dot{\mathbf{s}} \right]. \quad (56)$$

373 Substituting the cable equation (19) with $\mathbf{v}_q = \mathbf{e}_v$ and using $P_T(\mathbf{q})\mathbf{s} = \mathbf{s}$,
374 and $\mathbf{s}^\top \mathbf{q} = 0$ yields

$$\begin{aligned} \ell (\mathbf{q} + \mathbf{e}_3)^\top \dot{\mathbf{s}} &= -(\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) (\dot{\mathbf{e}}_v + g \mathbf{e}_3) - \frac{c_t}{m_L} (\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) (\mathbf{e}_v + \ell \mathbf{s}) \\ &\quad - \|\mathbf{s}\|^2 \underbrace{(\mathbf{q} + \mathbf{e}_3)^\top \mathbf{q}}_{= 1 + \mathbf{q}^\top \mathbf{e}_3 = \frac{1}{2} \|\mathbf{q} + \mathbf{e}_3\|^2}. \end{aligned} \quad (57)$$

375 Using

$$(\mathbf{q} + \mathbf{e}_3)^\top \mathbf{q} = 1 + \mathbf{q}^\top \mathbf{e}_3 = \frac{1}{2} \|\mathbf{q} + \mathbf{e}_3\|^2, \quad (58)$$

376 the last term becomes higher order near the equilibrium.

377 Next, the gravity-restoring term can be evaluated explicitly as follows.

378 Using the identity $P_T(\mathbf{q}) = I - \mathbf{q}\mathbf{q}^\top$, we obtain

$$(\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) g \mathbf{e}_3 = g (\mathbf{q} + \mathbf{e}_3)^\top (\mathbf{e}_3 - \mathbf{q}(\mathbf{q}^\top \mathbf{e}_3)). \quad (59)$$

379 Expanding the inner products and using $\mathbf{q}^\top \mathbf{q} = 1$ gives

$$(\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) g \mathbf{e}_3 = g(1 + \mathbf{q}^\top \mathbf{e}_3)(1 - \mathbf{q}^\top \mathbf{e}_3). \quad (60)$$

380 Since $\mathbf{q} \in S^2$, the identity $\|\mathbf{q} + \mathbf{e}_3\|^2 = 2(1 + \mathbf{q}^\top \mathbf{e}_3)$ holds, and therefore

$$(\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) g \mathbf{e}_3 = g \frac{\|\mathbf{q} + \mathbf{e}_3\|^2}{2} (1 - \mathbf{q}^\top \mathbf{e}_3). \quad (61)$$

381 In the hemisphere $\mathbf{q}^\top \mathbf{e}_3 \leq 0$ containing the hanging equilibrium $\mathbf{q} = -\mathbf{e}_3$,
382 we have

$$1 - \mathbf{q}^\top \mathbf{e}_3 \geq 1, \quad (62)$$

383 and hence

$$(\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) g \mathbf{e}_3 \geq \frac{g}{2} \|\mathbf{q} + \mathbf{e}_3\|^2. \quad (63)$$

384 This bound holds for all $\mathbf{q}$ in a neighbourhood of the equilibrium $\mathbf{q} = -\mathbf{e}_3$
385 contained in the hemisphere $\mathbf{q}^\top \mathbf{e}_3 \leq 0$.

386 Substituting (57) and (63) into (56), and noting that the $-\|\mathbf{s}\|^2 \|\mathbf{q} + \mathbf{e}_3\|^2 / (2\ell)$
387 term from (57) is $O(\|z\|^3)$ near the origin and absorbed into $\tilde{R}_c$ below:

$$\dot{V}_c \leq \underbrace{\eta m_L \ell^2 \|\mathbf{s}\|^2}_{(\star)} - \frac{\eta m_L g \ell}{2} \|\mathbf{q} + \mathbf{e}_3\|^2 + \eta m_L \ell^2 \tilde{R}_c, \quad (64)$$

388 where $\tilde{R}_c$ collects only higher-order contributions:

$$\tilde{R}_c := -\frac{1}{\ell} (\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) \dot{\mathbf{e}}_v - \frac{c_t}{m_L \ell} (\mathbf{q} + \mathbf{e}_3)^\top P_T(\mathbf{q}) (\mathbf{e}_v + \ell \mathbf{s}) - \frac{\|\mathbf{s}\|^2 \|\mathbf{q} + \mathbf{e}_3\|^2}{2\ell^2}. \quad (65)$$

389 We verify that each summand in $\tilde{R}_c$ is at most $O(\|z\|^2)$ near the equilib-
390 rium. Writing $\mathbf{q} = -\mathbf{e}_3 + \boldsymbol{\xi}$ with $\boldsymbol{\xi} := \mathbf{q} + \mathbf{e}_3$, the constraint $\mathbf{q}^\top \mathbf{s} = 0$ gives
391 $(-\mathbf{e}_3 + \boldsymbol{\xi})^\top \mathbf{s} = 0$, i.e. $\mathbf{e}_3^\top \mathbf{s} = \boldsymbol{\xi}^\top \mathbf{s}$; hence every inner product of $\mathbf{s}$ with a fixed
392 vector is $O(\|\boldsymbol{\xi}\| \|\mathbf{s}\|) = O(\|z\|^2)$. Using this together with $\|\boldsymbol{\xi}\| = O(\|z\|)$,
393 $\|\dot{\mathbf{e}}_v\| = O(\|z\|)$ (from (35b)), and $P_T(\mathbf{q}) \mathbf{s} = \mathbf{s}$:

394 $\bullet \frac{1}{\ell} \boldsymbol{\xi}^\top P_T \dot{\mathbf{e}}_v = O(\|\boldsymbol{\xi}\| \|\dot{\mathbf{e}}_v\|) = O(\|z\|^2);$

- 395 • $\frac{c_t}{m_L \ell} \boldsymbol{\xi}^\top P_T \mathbf{e}_v = O(\|\boldsymbol{\xi}\| \|\mathbf{e}_v\|) = O(\|z\|^2);$
- 396 • $\frac{c_t}{m_L} \boldsymbol{\xi}^\top P_T \mathbf{s} = \frac{c_t}{m_L} \boldsymbol{\xi}^\top \mathbf{s} = \frac{c_t}{m_L} \mathbf{e}_3^\top \mathbf{s} = O(\|z\|^2)$ (here the constraint $\mathbf{q}^\top \mathbf{s} = 0$
- 397 *is essential*);
- 398 • $\frac{1}{2\ell^2} \|\mathbf{s}\|^2 \|\boldsymbol{\xi}\|^2 = O(\|z\|^4).$

399 The first three terms are $O(\|z\|^2)$, whereas the last one is $O(\|z\|^4)$. Since the
 400 analysis is local, $O(\|z\|^4)$ can be absorbed into the quadratic bound. Hence,
 401 there exists $C_R > 0$ such that

$$|\tilde{R}_c| \leq C_R \|z\|^2 \quad (66)$$

402 for all z in a sufficiently small neighbourhood $\mathcal{U}$ of the equilibrium.

403 The positive term $(\star)$ is handled in the combining step below: adding
 404 $(\star)$ to the $-\gamma_s \|\mathbf{s}\|^2$ from $\dot{V}_2$ gives a net coefficient $-(\gamma_s - \eta m_L \ell^2) < 0$ since
 405 $\gamma_s - \eta m_L \ell^2 > 0$; (see (32f)).

406 **Step 5: combining.** Adding (48), (54), and (64), the combined derivative
 407 $\dot{V}$ (excluding the $\eta m_L \ell^2 C_R \|z\|^2$ remainder from Step 4) contains four cross
 408 terms in $\|\mathbf{s}\|$ that must be absorbed:

$$MK_c \|\mathbf{e}_v\| \|\mathbf{s}\|, \quad (67)$$

$$M\sigma K_c \|\mathbf{e}_p\| \|\mathbf{s}\|, \quad (68)$$

$$m_L \ell K_p \|\mathbf{e}_p\| \|\mathbf{s}\|, \quad (69)$$

$$\ell |m_L K_d - c_t| \|\mathbf{e}_v\| \|\mathbf{s}\|. \quad (70)$$

409 Applying Young's inequality $ab \leq \frac{\varepsilon}{2} a^2 + \frac{1}{2\varepsilon} b^2$ with $\varepsilon = \frac{\lambda_{\min}(A)}{2}$ to each term
 410 (the first two consume the $\|\mathbf{e}_v\|^2$ and $\|\mathbf{e}_p\|^2$ budgets from $\dot{V}_1$, the last two

411 from $\dot{V}_2$):

$$MK_c \|\mathbf{e}_v\| \|\mathbf{s}\| \leq \frac{\lambda_{\min}(A)}{4} \|\mathbf{e}_v\|^2 + \frac{M^2 K_c^2}{\lambda_{\min}(A)} \|\mathbf{s}\|^2, \quad (71)$$

$$M\sigma K_c \|\mathbf{e}_p\| \|\mathbf{s}\| \leq \frac{\lambda_{\min}(A)}{4} \|\mathbf{e}_p\|^2 + \frac{M^2 \sigma^2 K_c^2}{\lambda_{\min}(A)} \|\mathbf{s}\|^2, \quad (72)$$

$$m_L \ell K_p \|\mathbf{e}_p\| \|\mathbf{s}\| \leq \frac{\lambda_{\min}(A)}{4} \|\mathbf{e}_p\|^2 + \frac{m_L^2 \ell^2 K_p^2}{\lambda_{\min}(A)} \|\mathbf{s}\|^2, \quad (73)$$

$$\ell |m_L K_d - c_t| \|\mathbf{e}_v\| \|\mathbf{s}\| \leq \frac{\lambda_{\min}(A)}{4} \|\mathbf{e}_v\|^2 + \frac{\ell^2 (m_L K_d - c_t)^2}{\lambda_{\min}(A)} \|\mathbf{s}\|^2. \quad (74)$$

412 Summing (71)–(74) and collecting the $\|\mathbf{e}_v\|^2$, $\|\mathbf{e}_p\|^2$, and $\|\mathbf{s}\|^2$ terms yields

$$\dot{V} \leq -\frac{\lambda_{\min}(A)}{2} (\|\mathbf{e}_v\|^2 + \|\mathbf{e}_p\|^2) - \beta_s \|\mathbf{s}\|^2 - \frac{\eta m_L g \ell}{2} \|\mathbf{q} + \mathbf{e}_3\|^2 + \eta m_L \ell^2 C_R \|z\|^2, \quad (75)$$

413 where

$$\beta_s = \gamma_s - \eta m_L \ell^2 - \frac{M^2 K_c^2 (1 + \sigma^2) + m_L^2 \ell^2 K_p^2 + \ell^2 (m_L K_d - c_t)^2}{\lambda_{\min}(A)} > 0, \quad (76)$$

414 with

$$\gamma_s := c_t \ell^2 - m_L \ell K_c > 0. \quad (77)$$

415 The condition $\beta_s > 0$ is sufficient and may be conservative; in particular, the
 416 term $\ell^2 (m_L K_d - c_t)^2$ is minimized when $K_d = c_t / m_L$, which improves the
 417 feasibility of the estimate. Define

$$\beta_0 = \min\left(\frac{\lambda_{\min}(A)}{2}, \beta_s, \frac{\eta m_L g \ell}{2}\right) > 0. \quad (78)$$

418 Then (75) implies

$$\dot{V} \leq -(\beta_0 - \eta m_L \ell^2 C_R) \|z\|^2. \quad (79)$$

419 If

$$\beta_0 > \eta m_L \ell^2 C_R, \quad (80)$$

420 then there exists $c > 0$ such that

$$\dot{V} \leq -c \|z\|^2 \leq -\frac{c}{\alpha_2} V =: -\rho V, \quad (81)$$

421 for all z in a neighbourhood $\mathcal{U}$ of the equilibrium. This establishes local
422 exponential stability and bounds (i)–(ii).

423 **Step 6: cable tautness (iii).** At equilibrium, $\lambda^* = m_L g > 0$ by Lemma 2.
424 The tension (24) is continuous in z (under tangential drag, $\mathbf{q}^\top \mathbf{f}_{\text{drag}} = 0$). By
425 continuity, there exists $\delta > 0$ with $\|z\| < \delta \Rightarrow \lambda \geq m_L g/2 > 0$. Choose r
426 above also satisfying $\{V \leq r\} \subset \{\|z\| < \delta\}$; forward invariance of $\{V \leq r\}$
427 by (81) then gives $\lambda(t) \geq m_L g/2 > 0$ for all $t \geq 0$. $\square$

428 **Remark 6** (Conservativeness of gain conditions). *Conditions (32c)–(32d)*
429 *are sufficient, but not necessary, for stability. In particular, condition (32c)*
430 *imposes the upper bound $K_c < c_t \ell / m_L$, which originates from ensuring neg-*
431 *ativity of the pure quadratic term in $\|\mathbf{s}\|^2$ within the Lyapunov derivative.*

432 *Condition (32d) is more restrictive and reflects the use of Young’s in-*
433 *equality to absorb the bilinear cross terms $\mathbf{e}_p^\top \mathbf{s}$ and $\mathbf{e}_v^\top \mathbf{s}$ into quadratic terms.*
434 *This step introduces conservative bounds that must be dominated by γ_s .*

435 *The feasibility of (32d) improves by selecting $K_d \approx c_t / m_L$, which mini-*
436 *mizes the contribution of the term $\ell^2(m_L K_d - c_t)^2$ in β_s .*

437 *Therefore, conditions (32b)–(32d) provide a transparent analytical design*
438 *guideline ensuring local exponential stability, although stability may hold for*
439 *a broader set of gains than predicted by these sufficient conditions. Con-*
440 *sequently, the analytical gain inequalities should be interpreted as a suffi-*
441 *cient local certification tool rather than as a tuning rule for high-performance*
442 *trajectory-tracking simulations.*

443 **Remark 7** (Practical tracking for time-varying references). *Theorem 1 is*
 444 *stated for a constant reference $\mathbf{p}_d^*$. For a time-varying reference with $\dot{\mathbf{p}}_d, \ddot{\mathbf{p}}_d \in$*
 445 *L^∞ , the precompensated error dynamics (27) remain exact (the precompen-*
 446 *sator eliminates all projection residuals regardless of the reference), but ad-*
 447 *ditional terms appear in $\dot{V}$: the term $-c_t \ell \mathbf{s}^\top \dot{\mathbf{p}}_d$ in $\dot{V}_2$ and an extra $\ddot{\mathbf{p}}_d$ -*
 448 *dependent forcing in $\dot{\mathbf{e}}_v$. These terms prevent a clean convergence-to-zero*
 449 *conclusion. Moreover, for accelerating references the cable equilibrium is no*
 450 *longer the vertical configuration $\mathbf{q} = -\mathbf{e}_3$, so the pendulum energy compo-*
 451 *nent of the Lyapunov function loses its interpretation as a distance to the*
 452 *true equilibrium. A complete Lyapunov proof for arbitrary time-varying ref-*
 453 *erences is outside the scope of this paper. The figure-eight simulation of*
 454 *Section 4 instead evaluates the controller in a demanding practical track-*
 455 *ing scenario, illustrating that the stability basin is large enough for smooth*
 456 *trajectory-tracking tasks with $\sup_t \|\dot{\mathbf{p}}_d\| \approx 0.88$ m/s and $\sup_t \|\ddot{\mathbf{p}}_d\| \approx 0.55$*
 457 *m/s².*

458 **Remark 8** (Cascade stability). *When the inner attitude loop is included, the*
 459 *full system is a cascade of the exponentially stable attitude loop driving the*
 460 *exponentially stable outer loop (with the attitude error as a bounded, decaying*
 461 *input). By standard cascade stability results [28], the full closed-loop is locally*
 462 *exponentially stable.*

463 3.4. Operational tension characteristics

464 Theorem 1 guarantees $\lambda(t) > 0$ as a consequence of the exponential con-
 465 vergence to equilibrium (Objective O3), without any explicit tension refer-
 466 ence. Nevertheless, it is useful to characterise the *operating tension* during

467 the transient and in different flight regimes. Table 2 lists typical scenarios;
 468 in each case λ can be estimated online via (12) and used as a monitoring
 469 signal (e.g. to trigger a slack-cable contingency mode).

Table 2: Typical operating tension scenarios. In all cases the controller maintains $\lambda > 0$; no separate tension reference is tracked.

Scenario	Expected $\hat{\lambda}$	Monitoring action
Quasi-static hover	$m_L g$	Nominal
Figure-eight flight	$m_L g \pm \Delta_{\text{accel}}$	$\Delta_{\text{accel}} = m_L \ \ddot{\mathbf{p}}_d\ $
Aggressive maneuver	$> m_L g$ (centripetal)	Check thrust margin
Cable engagement	$0 \rightarrow m_L g$ (transient)	Slack monitor active

470 3.5. Implementation remarks

471 *Connection to geometric control.* The outer-loop control law (25) defines the
 472 desired thrust direction $\mathbf{b}_3^*$. This is consistent with the geometric control
 473 framework of [7, 8] and can be used directly as the thrust-vector reference
 474 for any SO(3)-based attitude controller.

475 *Numerical preservation of constraints.* The constraints $\|\mathbf{q}\| = 1$ and $\mathbf{q}^\top \mathbf{s} = 0$
 476 must be enforced at each integration step. Practical methods include pro-
 477 jecting after each Euler step ($\mathbf{q} \leftarrow \mathbf{q}/\|\mathbf{q}\|$, $\mathbf{s} \leftarrow P_T \mathbf{s}$) or using a geometric
 478 integrator on $T\mathbb{S}^2$.

479 *Cable-taut monitoring.* The taut condition $\lambda \geq 0$ is monitored online via
 480 (24). If $\lambda < 0$ is detected (slack cable), the trajectory should be slowed or a
 481 slack-cable extension with complementarity constraints should be activated.

482 *Physical thrust and attitude allocation.* Given $\mathbf{u}^*$ from (25): $T^* = m_q \|\mathbf{u}^*\|$,
 483 $\mathbf{b}_3^* = \mathbf{u}^* / \|\mathbf{u}^*\|$. The yaw reference $\psi_d(t)$ is set independently. When $\|\mathbf{u}^*\| \rightarrow$
 484 0, numerical regularisation is required.

485 *Integral action.* For robustness against constant drag or uncertain load mass,
 486 an integral term can be added to the position controller: $\mathbf{a}_{\text{track}} \rightarrow \mathbf{a}_{\text{track}} -$
 487 $K_i \int \mathbf{e}_p dt$, with anti-windup when thrust is saturated.

488 4. Simulation Results

489 The complete 24-state closed-loop system (21) under the cascade con-
 490 troller (25)–(31) is evaluated on a demanding figure-eight trajectory to assess
 491 the practical tracking capability of the proposed design beyond the constant-
 492 reference assumptions of Theorem 1.

493 4.1. Simulation setup

494 The parameter values are listed in Table 3. The desired trajectory $\mathbf{p}_d(t)$
 495 is a lemniscate of Bernoulli (figure-eight) at constant height $z_h = 1.5$ m,
 496 traversed with period $T_8 = 10$ s and horizontal amplitude $A = 1.4$ m:

$$\mathbf{p}_d(t) = S(t) \begin{bmatrix} \frac{A \cos \omega t}{1 + \sin^2 \omega t} \\ \frac{A \cos \omega t \sin \omega t}{1 + \sin^2 \omega t} \\ z_h \end{bmatrix}, \quad \omega = \frac{2\pi}{T_8}, \quad (82)$$

497 where $S(t) = 10\tau^3 - 15\tau^4 + 6\tau^5$ with $\tau = \min(t/T_r, 1)$ and $T_r = 2.5$ s is a
 498 quintic envelope that smoothly ramps the trajectory from rest.

499 The initial condition places the drone at rest at $\mathbf{p}_q(0) = (0, 0, 1.5)^\top$, with
 500 the cable deflected 20 from vertical ($\mathbf{q}(0) = [\sin 20 \cos 30, \sin 20 \sin 30, -\cos 20]^\top$)

501 and the drone levelled ($R(0) = I$, $\Omega(0) = \mathbf{0}$). The equations of motion were
 502 integrated with an adaptive-step Runge–Kutta 4(5) solver (relative tolerance
 503 10^{-5} , absolute tolerance 10^{-7}), with SO(3)-drift corrected at each output
 504 step via SVD re-orthogonalization and the unit-sphere constraint enforced
 505 by renormalizing $\mathbf{q}$.

506 The desired attitude $R_d(t)$ required by the inner loop was precomputed
 507 on a fine grid from the outer-loop ideal trajectory (Section 3.2), avoiding nu-
 508 merical stiffness in the coupled 24-state ODE. All control gains are collected
 509 in Table 3.

510 The simulation gains $K_p = 45$, $K_d = 12$, $K_c = 0.5$, $K_R = 6$, $K_\Omega = 2$,
 511 with system parameters $m_L = 0.35$ kg, $\ell = 0.80$ m, and $c_t = 0.25$, satisfy the
 512 basic cable-damping condition

$$\gamma_s = c_t \ell^2 - m_L \ell K_c = 0.25(0.8)^2 - 0.35(0.8)(0.5) = 0.020 > 0.$$

513 The remaining inequalities in Theorem 1 are sufficient and conservative; the
 514 present figure-eight simulation uses more aggressive, performance-oriented
 515 gains and is therefore reported as a numerical trajectory-tracking test beyond
 516 the certified constant-reference gain set.

517 **Remark 9** (Open-source code and supplementary video). *The complete*
 518 *Python simulation code, including the 3D animation, numerical integrator,*
 519 *and figure-generation scripts, is publicly available at*

520 *<https://github.com/gfloresc/quadrotor-suspended-load-control>*

521 *The repository also contains supplementary videos of the closed-loop simula-*
 522 *tion demonstrating the figure-eight maneuver, cable-swing attenuation, and*
 523 *tension evolution in real time.*

Table 3: System parameters and control gains used in simulation.

Symbol	Value	Symbol	Value
m_q	1.20 kg	K_p	45.0 s ⁻²
m_L	0.35 kg	K_d	12.0 s ⁻¹
ℓ	0.80 m	K_c	0.5
$J_{xx} = J_{yy}$	1.5×10^{-2} kg m ²	K_R	6.0 N m
J_{zz}	2.5×10^{-2} kg m ²	K_Ω	2.0 N m s
c_t	0.25 N s/m	g	9.81 m/s ²

524 4.2. Results

525 Figure 3 shows a rendered simulation snapshot at $t = 18.50$ s, midway
526 through the figure-eight maneuver. The drone body, cable, and suspended
527 load are rendered in three dimensions alongside the live plots of position
528 error, cable swing, tension, and attitude error; the thrust vector $\mathbf{b}_3$ (orange)
529 and desired direction $\mathbf{b}_3^*$ (purple dashed) are nearly coincident, confirming
530 the fast inner-loop convergence discussed in Section 3.2.

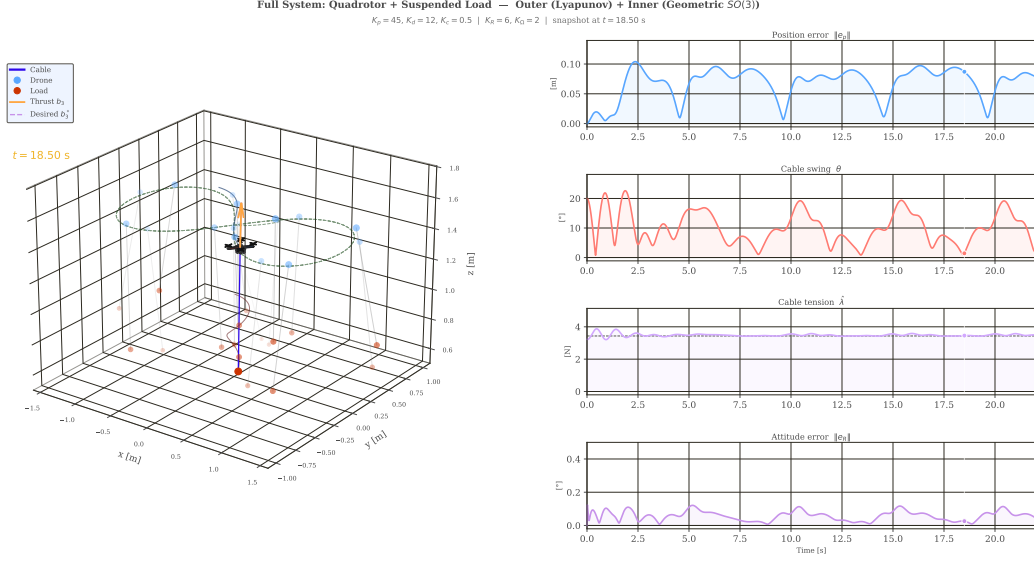


Figure 3: Rendered composite view of the closed-loop simulation at $t = 18.50$ s. *Left*: three-dimensional view of the quadrotor (black), cable (blue), and load (red dot) mid figure-eight, with the instantaneous thrust direction $\mathbf{b}_3$ (orange arrow) and desired $\mathbf{b}_3^*$ (purple dashed arrow) nearly overlapping. *Right*: live time-history plots of position error $\|e_p\|$, cable swing θ , tension $\hat{\lambda}$, and attitude error $\|e_R\|$, with a vertical cursor at the current time.

Figure 4 shows the three-dimensional trajectories of the drone, reference, and load over the full 22 s simulation. The drone converges to the figure-eight reference within the ramp interval and maintains tracking thereafter. The cable-suspended load traces a figure-eight at altitude $z_L \approx z_h - \ell \cos \theta$, displaced vertically below the drone; the instantaneous cable configurations (grey lines) confirm that the cable remains taut throughout the maneuver.

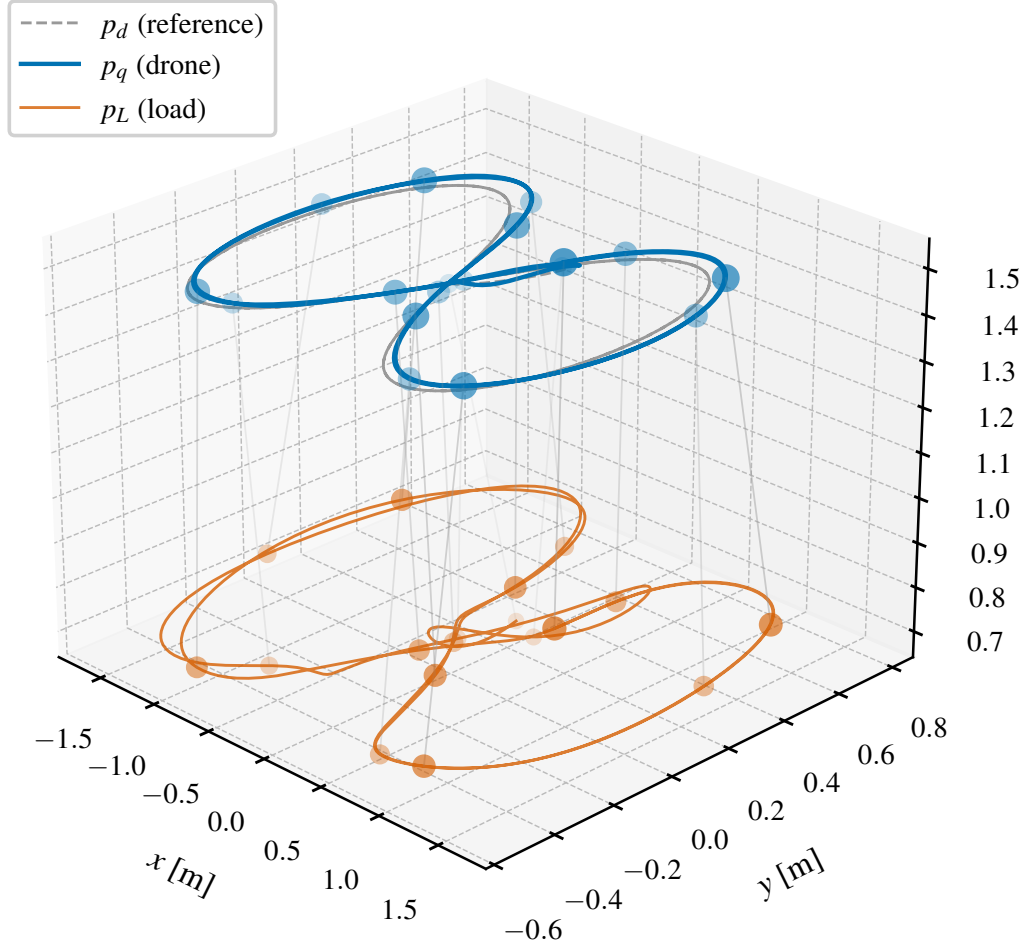


Figure 4: Three-dimensional trajectories of drone p_q (blue), load p_L (orange), and reference p_d (dashed grey). Grey lines indicate instantaneous cable configurations at uniform time intervals; marker size increases with time to indicate temporal ordering.

Figure 5 presents the tracking and cable-dynamics performance. Panel (a) shows the three components of the position error $\mathbf{e}_p = \mathbf{p}_q - \mathbf{p}_d$ together with $\|\mathbf{e}_p\|$: the peak error is 0.104 m during the initial transient, the final error at $t = 22$ s is 0.079 m, and the RMS error over the last figure-eight cycle is 0.076 m, confirming sustained bounded tracking throughout the maneuver.

542 Panel (b) shows the velocity error $\mathbf{e}_v = \mathbf{v}_q - \mathbf{v}_d$, which converges faster than
 543 the position error.

544 Panel (c) shows the cable cone angle $\theta(t) = \arccos(-\mathbf{q}^\top \mathbf{e}_3)$ in degrees,
 545 measuring total deflection of the load from the hanging vertical. Starting
 546 from $\theta_0 = 20$, the angle rises briefly to a peak of 22.62 during the initial
 547 transient as the drone accelerates toward the reference, then is progressively
 548 attenuated by the active damping term $-K_c P_T(\mathbf{q})\mathbf{s}$ in (25), reaching $\theta(22) =$
 549 2.49 at the end of the simulation. The RMS swing over the last figure-eight
 550 cycle is 10.60, reflecting the bounded oscillations induced by the periodic
 551 reference during practical tracking; convergence to zero is the theoretical
 552 guarantee only for the constant-reference case of Theorem 1.

553 Panel (d) confirms O3 (cable tautness): the minimum estimated tension
 554 over the entire simulation is $\hat{\lambda}_{\min} = 3.207$ N, which is 93.4% of the nominal
 555 hover tension $m_L g = 3.434$ N. The tension never approaches zero, and con-
 556 verges to 3.494 N at $t = 22$ s, slightly above $m_L g$ due to residual centripetal
 557 loading from the periodic trajectory. This verifies that the radial thrust chan-
 558 nel maintains cable tautness as a consequence of the correct hover equilibrium
 559 $(1 + \mu)g\mathbf{e}_3$ embedded in (25).

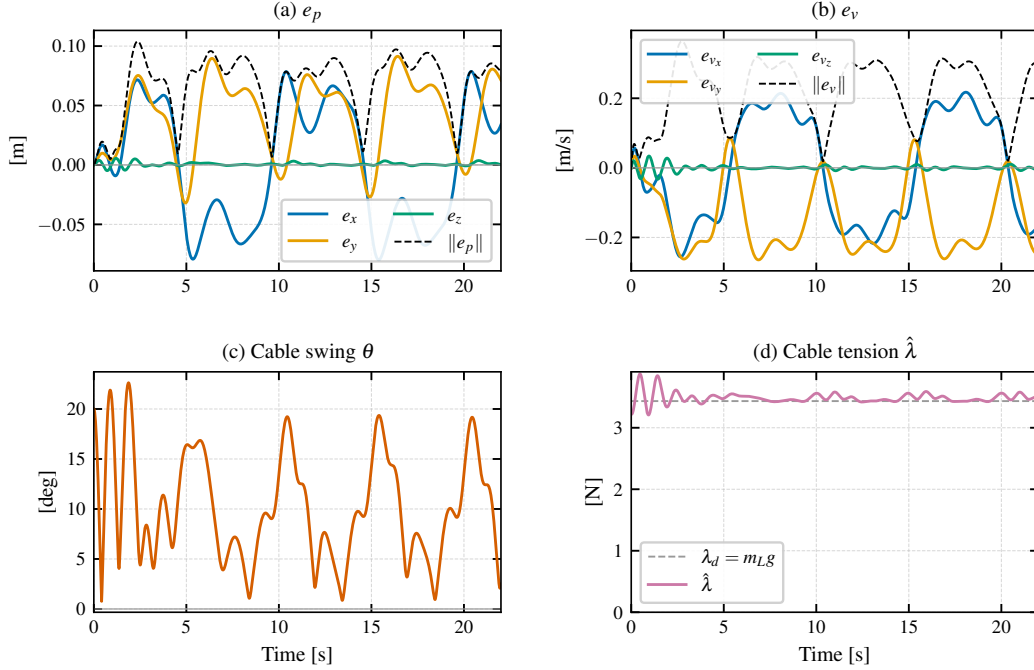


Figure 5: Tracking and cable dynamics. (a) Position error components and norm $\|e_p\|$. (b) Velocity error components and norm $\|e_v\|$. (c) Cable swing angle $\theta(t) = \arccos(-\mathbf{q}^\top \mathbf{e}_3)$, the *cone angle* measuring total deflection of the cable from the hanging vertical. Because the load moves on $\mathbb{S}^2$, its configuration requires two angles (colatitude θ and azimuth ϕ); only θ is shown here since it directly quantifies deviation from equilibrium and corresponds to the Lyapunov bound $\|\mathbf{q} + \mathbf{e}_3\| \approx \sin \theta$ (Objective O2). The azimuth ϕ is unobservable from θ alone but does not affect tautness or tracking performance. (d) Estimated cable tension $\hat{\lambda}$ versus the equilibrium value $\lambda_d = m_L g$ (dashed).

560 *Inner-loop performance.* Figure 6 shows the attitude and angular-velocity
 561 tracking errors of the inner geometric loop. The attitude error $\|e_R\|$ (panel (a))
 562 reaches a peak of 0.12 during the initial transient and decays below 0.05
 563 within 3 s. The angular-velocity error $\|e_\Omega\|$ (panel (b)) follows a similar pro-
 564 file. Both errors are an order of magnitude smaller than the corresponding
 565 outer-loop errors during steady-state tracking, confirming the time-scale sep-

566 aration assumed in the cascade stability argument of Remark 8: the inner
 567 loop is effectively faster than the outer loop, so Assumption 4 is well-justified
 568 at these gain values.

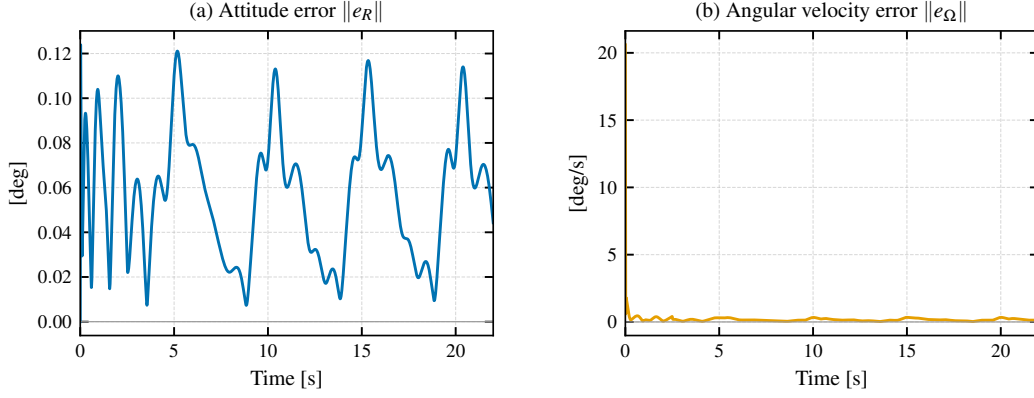


Figure 6: Inner-loop performance. (a) Attitude error $\|e_R\|$. (b) Angular-velocity error $\|e_\Omega\|$. Both quantities converge an order of magnitude faster than the outer-loop errors, validating the cascade separation.

569 *Control signals.* Figure 7 examines the control signals. Panel (a) shows the
 570 total thrust $T(t)$ alongside the hover reference $(m_q + m_L)g = 15.2$ N (dashed):
 571 the thrust remains positive throughout, oscillates by at most ± 0.5 N around
 572 the hover value during steady-state tracking, and never saturates, confirming
 573 physical feasibility.

574 Panel (b) shows $\|\tau\|$, which peaks at 0.4 N m during the initial atti-
 575 tude correction and decays rapidly; this is consistent with the fast inner-loop
 576 convergence observed in Fig. 6. Panel (c) shows the individual torque com-
 577 ponents τ_x, τ_y, τ_z : all three remain below 0.05 N m in steady state, well within
 578 typical actuator limits.

579 Panel (d) shows the three components of the precompensated virtual

control $\mathbf{u}^* = (I + \mu \mathbf{q} \mathbf{q}^\top)(g \mathbf{e}_3 + \mathbf{a}_d - K_p \mathbf{e}_p - K_d \mathbf{e}_v - K_c \mathbf{s}) - \mu \ell \|\mathbf{s}\|^2 \mathbf{q}$ from
(25). At steady state $\mathbf{q} \approx -\mathbf{e}_3$, the precompensator evaluates to $(I + \mu \mathbf{e}_3 \mathbf{e}_3^\top)$,
so $u_z^* \approx (1 + \mu)g = 12.7 \text{ m/s}^2$ (dominated by gravity compensation) remains
nearly constant, while u_x^* and u_y^* carry tracking and damping corrections.
This confirms the geometric decoupling of Proposition 1: the radial channel
(u_z^*) and the tangential channel (u_x^*, u_y^*) operate largely independently.

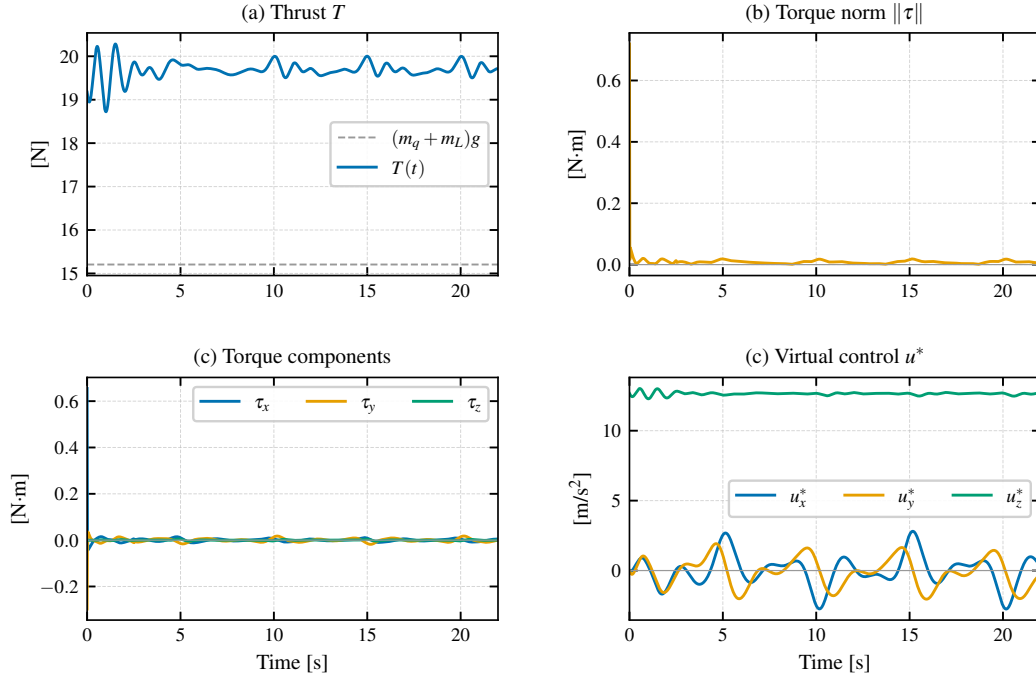


Figure 7: Control signals. (a) Total thrust $T(t)$ versus hover reference $(m_q + m_L)g$ (dashed). (b) Torque norm $\|\tau\|$. (c) Individual torque components τ_x, τ_y, τ_z . (d) Virtual control vector $\mathbf{u}^*$: the z -component dominates (gravity compensation) while x and y carry the tracking and damping corrections, illustrating the radial-tangential decoupling.

Quantitative summary. Table 4 collects the key performance metrics.

Table 4: Quantitative simulation results for the figure-eight trajectory (practical tracking scenario, $T_{\text{sim}} = 22$ s).

Metric	Value	Objective
Peak position error $\max \ \mathbf{e}_p\ $	0.104 m	O1 (position)
Final position error $\ \mathbf{e}_p(22)\ $	0.079 m	O1 (position)
RMS position error (last cycle)	0.076 m	O1 (position)
Initial cable swing $\theta(0)$	20.0	O2 (swing)
Peak cable swing $\max \theta$	22.62	O2 (swing)
Final cable swing $\theta(22)$	2.49	O2 (swing)
RMS cable swing (last cycle)	10.60	O2 (swing)
Minimum cable tension $\min \hat{\lambda}$	3.207 N	O3 (tautness)
Final cable tension $\hat{\lambda}(22)$	3.494 N	O3 (tautness)
Nominal hover tension $m_L g$	3.434 N	O3 (tautness)
Peak attitude error $\max \ \mathbf{e}_R\ $	0.124	Inner loop

587 4.3. Practical sensing considerations

588 The simulation above assumes full-state feedback, with the cable states
589 $(\mathbf{q}, \mathbf{s})$ available directly. In practice, these states may be estimated using a
590 load-mounted IMU, motion-capture measurements, or sensor fusion between
591 the drone and load states. As one possible reconstruction route: given the
592 load position $\mathbf{p}_L$ and velocity $\mathbf{v}_L$ (from GPS, barometer, or IMU integration),
593 the cable direction and angular velocity follow from

$$\hat{\mathbf{q}} = \frac{\mathbf{p}_L - \mathbf{p}_q}{\ell}, \quad \hat{\mathbf{s}} = \frac{\mathbf{v}_L - \mathbf{v}_q}{\ell}, \quad (83)$$

594 with renormalization $\hat{\mathbf{q}} \leftarrow \hat{\mathbf{q}}/\|\hat{\mathbf{q}}\|$ and tangential projection $\hat{\mathbf{s}} \leftarrow (I - \hat{\mathbf{q}}\hat{\mathbf{q}}^\top)\hat{\mathbf{s}}$
 595 enforced at each control step to preserve the unit-sphere constraint. A com-
 596 plete observer design accounting for accelerometer bias, integration drift,
 597 sensor orientation, and colored measurement noise, as well as noise-robust
 598 experimental validation, are beyond the scope of the present paper and are
 599 identified as directions for future work (Section 5).

600 5. Conclusions

601 This paper developed a complete nonlinear control framework for a quadro-
 602 tor with a cable-suspended load on the full configuration manifold $\mathbb{R}^3 \times$
 603 $\text{SO}(3) \times T\mathbb{S}^2$. The central result is a geometrically exact tangential–radial de-
 604 composition of the drone thrust vector (Proposition 1), which shows that po-
 605 sition tracking and cable-tension regulation occupy *independent* scalar chan-
 606 nels — a fact that follows directly from the unit-sphere geometry of $\mathbf{q} \in \mathbb{S}^2$
 607 and does not rely on any approximation or linearization.

608 Exploiting this structure, the outer-loop control law

$$\mathbf{u}^* = (I + \mu\mathbf{q}\mathbf{q}^\top)(g\mathbf{e}_3 + \mathbf{a}_d - K_p\mathbf{e}_p - K_d\mathbf{e}_v - K_c\mathbf{s}) - \mu\ell\|\mathbf{s}\|^2\mathbf{q}$$

609 embeds four design insights simultaneously: (i) the precompensator $(I +$
 610 $\mu\mathbf{q}\mathbf{q}^\top) = \Sigma(\mathbf{q})^{-1}$ exactly cancels the Sherman–Morrison operator, yielding
 611 exact error dynamics $\dot{\mathbf{e}}_v = -K_p\mathbf{e}_p - K_d\mathbf{e}_v - K_c\mathbf{s}$ with no projection residuals;
 612 (ii) at equilibrium the law reduces to $(1 + \mu)g\mathbf{e}_3$, placing $\lambda = m_L g > 0$
 613 exactly; (iii) the centripetal term $-\mu\ell\|\mathbf{s}\|^2\mathbf{q}$ cancels the radial centripetal
 614 forcing, completing the exact inversion; (iv) $-K_c\mathbf{s}$ drives the cable angular
 615 velocity to zero through the augmented Lyapunov cross-coupling $m_L\ell\mathbf{e}_v^\top\mathbf{s}$.

616 Combined with the geometric $\text{SO}(3)$ inner loop of [2], the closed-loop system
 617 has 24 states with no representational singularities.

618 Theorem 1 establishes local exponential stability using a structured Lya-
 619 punov certificate composed of position-error energy and spherical-pendulum
 620 mechanical energy, augmented with a coupling term that captures the inter-
 621 action between vehicle motion and cable deflection. The theoretical guar-
 622 antees are established for the constant-reference regulation problem, which
 623 captures the fundamental stabilization and swing-suppression mechanism,
 624 with fully explicit and verifiable gain conditions (32a)–(32d). The stability
 625 result is local because the pendulum energy component is positive definite
 626 only for cable deflections within the hemisphere about the downward equi-
 627 librium.

628 Numerical simulation on a lemniscate figure-eight trajectory ($A = 1.4$
 629 m, $T_8 = 10$ s) with a 20° initial cable deflection demonstrates that the
 630 same controller structure provides effective practical tracking beyond the
 631 constant-reference case: the peak position error was 0.104 m with a final
 632 value of 0.079 m, cable swing was attenuated from 20° to 2.49° at $t = 22$ s
 633 while remaining bounded during periodic tracking (RMS 10.60° over the last
 634 cycle), attitude error remained below 0.124° throughout, and cable tension
 635 stayed strictly positive with a minimum of 3.207 N (93.4% of the nominal
 636 hover tension $m_L g = 3.434$ N). The time-scale separation between the inner
 637 ($\|\mathbf{e}_R\|_\infty < 0.124^\circ$) and outer loops confirmed the cascade stability hypothesis
 638 and justified the perfect-attitude assumption used in the Lyapunov analysis.

639 Compared with the state of the art [19, 14, 17], the proposed approach
 640 is distinguished by three features not found jointly in any prior design: the

641 exact decoupling result, the provable tautness guarantee, and the singularity-
642 free formulation.

643 *Future work.* Open directions include: (a) semi-global or global stability via
644 energy-shaping methods for large initial swing angles $\theta_0 > 90^\circ$; (b) exper-
645 imental validation on a physical platform equipped with a motion-capture
646 system or a load-mounted IMU; (c) extension to teams of quadrotors sharing
647 a single payload, where the tension channel of each vehicle must be coordi-
648 nated to maintain a common cable-configuration equilibrium.

649 A particularly natural extension concerns *active tension regulation* for
650 scenarios in which the nominal value $\lambda = m_L g$ is no longer sufficient: vari-
651 able payload mass (e.g., progressive release of a water-bombing load), aggres-
652 sive maneuvers that risk momentary slack, aerodynamic drag on the cable,
653 or closed-loop positioning of the load tip in an aerial manipulation task.
654 The tangential–radial decoupling of Proposition 1 makes such an extension
655 straightforward. Because $\mathbf{q}^\top \mathbf{u}$ is the sole scalar that appears in the tension
656 formula (12), one can augment the control law with a desired radial compo-
657 nent $\alpha \in \mathbb{R}$ acting along $\mathbf{q}$:

$$\mathbf{u} = g\mathbf{e}_3 + \mathbf{a}_T + \alpha \mathbf{q}, \quad \mathbf{a}_T = P_T(\mathbf{a}_d - K_p \mathbf{e}_p - K_d \mathbf{e}_v), \quad (84)$$

658 where $P_T = I - \mathbf{q}\mathbf{q}^\top$ so that $\mathbf{q}^\top \mathbf{a}_T = 0$ exactly and the tangential channel is
659 undisturbed. Choosing

$$\alpha = \alpha_{\text{ff}} + k_\lambda(\lambda_d - \hat{\lambda}), \quad \alpha_{\text{ff}} = \ell \|\mathbf{s}\|^2 - \frac{1+\mu}{m_L}(\lambda_d - \mathbf{q}^\top \mathbf{f}_{\text{drag}}) - \mathbf{q}^\top g\mathbf{e}_3, \quad (85)$$

660 where $\hat{\lambda} = -\frac{m_L}{1+\mu}(\mathbf{q}^\top \mathbf{u} - \ell \|\mathbf{s}\|^2) + \mathbf{q}^\top \mathbf{f}_{\text{drag}}$ is the real-time tension estimate and
661 $k_\lambda > 0$ is a proportional feedback gain, yields $\lambda \rightarrow \lambda_d$ when the estimation

is consistent ($\hat{\lambda} \approx \lambda$). A desired tension profile $\lambda_d(t) \geq 0$ can be shaped with a quintic C^2 ramp between λ_0 and λ_1 to avoid exciting the pendulum mode during payload engagement or release. A full stability analysis of the augmented system (84)–(85), including the coupled tension–swing dynamics, is left for future work.

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